



**COMSATS University Islamabad,
Sahiwal Campus, Pakistan**

E-Commerce Website with AI-Powered

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of the requirement for the degree of**

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Executive Summary

The E-Commerce Website with AI-Powered Features is a modern, full-stack online shopping platform that integrates traditional electronic commerce workflows with advanced artificial intelligence capabilities to offer a more intelligent, personalized, and efficient buying experience to customers. The system addresses the widening gap between conventional e-commerce stores and the rapidly evolving expectations of online shoppers by bringing recommendation engines, intelligent search, personalization, and conversational chatbots into a single unified platform. By consolidating these capabilities, the project delivers a complete shopping ecosystem in which customers can browse products, receive intelligent suggestions, interact with an AI-driven support assistant, and complete secure transactions without leaving the website.

The platform is engineered using the MERN technology stack, comprising MongoDB as the database, Express.js as the backend framework, React.js as the frontend library, and Node.js as the runtime environment. This combination allows the system to deliver responsive user experiences, scalable backend services, and a flexible document-based storage layer capable of handling diverse product catalogues, customer profiles, orders, reviews, and machine-learning training data. The architecture is designed to follow a layered, service-oriented pattern in which the frontend communicates with the backend through well-defined REST APIs, ensuring clean separation of concerns, maintainability, and future extensibility.

Customers can register, sign in securely through JWT-based authentication, and browse through a rich catalogue of products organized by categories, brands, and attributes. The AI-based recommendation system analyses browsing patterns, purchase history, cart activity, and product similarity to suggest items that are most likely to interest each individual user. The smart search module uses natural language processing techniques to interpret user queries beyond simple keyword matching, enabling the system to understand intent, correct spelling errors, detect synonyms, and rank results based on relevance and personalization signals. The integrated chatbot offers round-the-clock customer support for common questions about products, orders, delivery, refunds, and account management, reducing the workload on human support teams and improving response times.

The administrative side of the platform is managed through a dedicated admin dashboard that provides tools for managing products, categories, orders, customers, payments, and overall business analytics. Administrators can monitor key performance indicators such as sales volume, revenue trends, popular products, conversion rates, and customer behavior patterns. The dashboard also includes inventory management, shipment tracking, and user role management, allowing the platform to be operated as a complete digital retail business.

Overall, the E-Commerce Website with AI-Powered Features represents a modern, scalable, and intelligent solution that bridges the gap between conventional online stores and next-generation intelligent commerce platforms. By offering secure authentication, personalized recommendations, intelligent search, conversational support, and comprehensive administrative tooling, the project delivers an end-to-end shopping ecosystem that can be adapted for real-world business deployment, research extension, and future academic study.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
DOM	Document Object Model
E-Commerce	Electronic Commerce
FR	Functional Requirements
HTML	Hypertext Markup Language
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
KNN	K-Nearest Neighbors
ML	Machine Learning

MERN	MongoDB, Express.js, React.js, Node.js
MVC	Model-View-Controller
NFR	Non-Functional Requirements
NLP	Natural Language Processing
NoSQL	Not Only SQL
OTP	One-Time Password
REST	Representative State Transfer
SDLC	Software Development Life Cycle
SEO	Search Engine Optimization
SPA	Single Page Application
SQL	Structured Query Language
SSL	Secure Sockets Layer
TLS	Transport Layer Security
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
XSS	Cross-Site Scripting

Table of Contents

1. Introduction.....	2
1.1 Brief	3
1.1.1 Purpose of the Project	3
1.1.2 Scope of the System.....	4
1.1.3 Technologies Used.....	4
1.1.4 System Overview	4
1.1.5 Outcome of the Project	5
1.2 Relevance to Course Modules	5
1.2.1 Software Engineering.....	5
1.2.2 Database Systems.....	5
1.2.3 Web Engineering and Internet Programming	6
1.2.4 Mobile Application Development.....	6
1.2.5 Object Oriented Programming.....	6
1.2.6 Data Structures and Algorithms.....	6
1.2.7 Human Computer Interaction	6
1.2.8 Information Security	7
1.2.9 Artificial Intelligence and Machine Learning.....	7
1.2.10 Final Year Project I and II	7
1.3 Project Background.....	7
1.4 Literature Review.....	8
1.5 Analysis from Literature Review.....	9
1.5.1 Alignment with Digital Commerce Adoption Trends	9
1.5.2 Comparison with Cloud Based E-Commerce Research	10
1.5.3 Relevance to Existing Recommendation and Search Systems	10
1.5.4 Integration of AI-Powered Features.....	10
1.5.5 Application of Security and Privacy Standards	10
1.5.6 Involvement of Mobile Commerce Concepts	10
1.5.7 Addressing Limitations of Existing E-Commerce Technologies	11
1.6 Methodology and Software Lifecycle.....	11
2. Problem Definition.....	14
2.1 Problem Statement	14
2.2 Deliverables and Development Requirements.....	14
2.3 Current System.....	16

2.4 Summary	17
3. Requirement Analysis	19
3.1 Introduction to Requirement Analysis	19
3.2 System Overview	19
3.3 Actors in the System	19
3.4 Use Case Diagrams	20
3.5 Use Case Descriptions	21
3.6 Functional Requirements	31
3.7 Non-Functional Requirements	33
4. Design and Architecture	36
4.1 System Architecture	36
4.1.1 Architecture Style	36
4.1.2 Components	36
4.1.3 Interactions	37
4.1.4 System Architecture Diagram	38
4.2 Data Representation	38
4.2.1 Database Design	38
4.2.2 Entity-Relationship Representation	39
4.2.3 Data Flow	40
4.2.4 Entity Descriptions and Constraints	42
4.3 Process Flow / Representation	42
4.3.1 Login and Authentication Process	42
4.3.2 Product Browsing and Search Process	42
4.3.3 Recommendation Generation Process	42
4.3.4 Payment Processing Workflow	42
4.4 Design Models	43
4.4.1 Class Model Description	43
4.4.2 Sequence Model Description	43
4.4.3 Activity Model Description	43
4.4.4 Component Model Description	43
5. Implementation	45
5.1 Algorithms	45
5.1.1 Authentication Algorithm	45
5.1.2 Recommendation Algorithm	45

5.1.3 Smart Search Algorithm	45
5.1.4 Chatbot Algorithm	46
5.1.5 Payment Processing Algorithm.....	46
5.2 External APIs	46
5.3 User Interface and Screenshots	47
5.3.1 Home Page	47
5.3.2 Login / Register Page.....	48
5.3.3 Product Listing Page	48
5.3.4 Product Detail Page.....	49
5.3.5 Shopping Cart Page.....	50
5.3.6 Checkout Page	51
5.3.7 Admin Dashboard	51
5.3.8 Admin — Product Management	52
5.3.9 Admin — Orders Management.....	52
6. Testing and Evaluation	54
6.1 Manual Testing	54
6.1.1 System Testing.....	54
6.1.2 Unit Testing	55
6.2 Automated Testing.....	57
6.3 Visual Test Evidence (Screenshots).....	59
6.3.1 Successful Login.....	59
6.3.2 Add to Cart.....	59
6.3.3 Order Placement.....	60
6.3.4 Error Handling	60
7. Conclusion	62
7.1 Future Work	62
7.1.1 Mobile Application	63
7.1.2 Multi-Vendor Marketplace	63
7.1.3 Subscription and Loyalty Programs.....	63
7.1.4 Advanced Analytics and Reporting	63
7.1.5 Integration with Voice Assistants and AR.....	63
7.1.6 Enhanced Security Features	63
8. References	65

List of Figures

Figure 1 use Case diagram of E-Commerce system	21
Figure 2: System Architecture Diagram	38
Figure 3: Entity-Relationship Diagram of the System.....	39
Figure 4: Detailed Document Relationship Structure in MongoDB.....	39
Figure 5: Sequence Diagram.....	40
Figure 6: Vertical Customer-Journey Activity Diagram	41
Figure 7: Home Page of E-Commerce Website.....	47
Figure 8: Login / Register Page of E-Commerce Website	48
Figure 9: Product Listing Page of E-Commerce Website.....	49
Figure 10: Product Detail Page of E-Commerce Website	49
Figure 11: Shopping Cart Page of E-Commerce Website	50
Figure 12: Checkout Page of E-Commerce Website	51
Figure 13: Admin Dashboard of E-Commerce Website.....	51
Figure 14: Admin Product Management of E-Commerce Website.....	52
Figure 15: Admin Orders Management of E-Commerce Website	52
Figure 16: Successful Login of E-Commerce Website.....	59
Figure 17: Add to Cart Functionality of E-Commerce Website	60
Figure 18: Error Handling Screens of E-Commerce Website	60

List of Tables

Table 1:Comparison of Existing E-Commerce Applications and the Proposed System	9
Table 2:Structured Description of the Login/Register Use Case	21
Table 3:Structured Description of the Set Profile Use Case	22
Table 4:Browse Products Use Case	23
Table 5:Smart Search Use Case	23
Table 6:Add to Cart Use Case	24
Table 7:Checkout and Payment Use Case	25
Table 8:View Recommendations Use Case	25
Table 9:Chatbot Interaction Use Case	26
Table 10:Review and Rate Product Use Case	27
Table 11:Manage Products (Admin) Use Case	28
Table 12:Manage Orders (Admin) Use Case	29
Table 13:Manage Users (Admin) Use Case	29
Table 14:View Analytics (Admin) Use Case	30
Table 15:Approve Seller Registration Use Case	31
Table 16:Details of APIs Used in the Project	46
Table 17:Unit Testing Table	55
Table 18:Functional Testing Table	56
Table 19:Integration Testing Table	57
Table 20:Automated Testing Table	58

Chapter 1

INTRODUCTION

1. Introduction

Electronic commerce has emerged as one of the most transformative forces in the modern global economy, reshaping the way goods are marketed, discovered, purchased, delivered, and consumed. Over the past two decades, e-commerce has evolved from simple online catalogues into complex ecosystems that incorporate recommendation engines, intelligent search, logistics tracking, digital payments, and automated customer support. In Pakistan, the growth of internet penetration, smartphone adoption, and digital payment services such as Easypaisa, Jazz Cash, and a growing number of bank gateways has created fertile ground for online shopping platforms to flourish. However, despite the rapid expansion of digital commerce, a significant number of local online stores still rely on traditional website structures that offer limited personalization, outdated search functionality, and inadequate support for intelligent decision-making. As a result, many customers have trouble in finding relevant products, are exposed to irrelevant promotions, and spend considerable time navigating through fragmented catalogues before making informed purchase decisions.

The traditional e-commerce experience in Pakistan, and across many emerging markets, often suffers from three interconnected problems. First, the product discovery process is largely dependent on rigid keyword-based search and basic category browsing, which fails to handle typographical errors, synonyms, or intent-based queries. Second, the recommendation mechanisms used in many existing platforms are either absent or limited to simple logic such as "most viewed" or "recently added", which ignores the individual preferences and browsing behaviors of the user. Third, customer support is predominantly manual, conducted through call centers or email, leading to long response times, inconsistent quality of service, and increased operational costs for store owners. These issues collectively reduce the quality of the user experience, lower conversion rates, and hamper the growth of digital commerce in the region.

In recent years, artificial intelligence has proven to be a powerful enabler for overcoming the limitations of conventional e-commerce platforms. Machine learning algorithms can analyze vast quantities of user behaviors data to generate personalized product recommendations; natural language processing techniques allow search systems to interpret the meaning behind a query rather than simply matching keywords; and conversational chatbots can handle large volumes of routine customer interactions while intelligently escalating complex cases to human agents. When these AI-driven capabilities are integrated within a well-designed online store, they transform the shopping journey into a personalized, efficient, and intelligent experience that can rival the quality of service offered by leading global platforms.

The E-Commerce Website with AI-Powered Features is developed as a modern solution to these challenges by combining the strengths of the MERN technology stack with carefully selected AI components. The platform is designed as a comprehensive digital commerce system that integrates traditional online shopping functionalities such as browsing, filtering, cart management, checkout, and order tracking with advanced AI capabilities including personalized recommendations, intelligent search, and a conversational chatbot. The system is architected around a clean separation of concerns, where the React.js frontend presents a rich and responsive user interface, the Node.js and Express.js backend exposes a set of secure REST APIs, and the MongoDB database stores product catalogues, user profiles, orders, reviews, and AI-related data structures. This foundation allows the system to serve a wide variety of users with fast, reliable, and scalable experience.

Customers of the platform can register, sign in using secure JWT-based authentication, browse through an organized catalogue of products, and filter results based on category, price, brand, rating, and other attributes. The AI-based recommendation engine analyses browsing activity, cart contents, purchase history, and product similarity to present each user with a personalized list of suggestions that align with their interests. The smart search module leverages natural language processing to interpret queries with greater accuracy, handle spelling mistakes, detect intent, and rank results with personalized relevance. The conversational chatbot is integrated across the platform to provide immediate answers to common questions about products, orders, delivery, and account management, thereby enhancing customer satisfaction and reducing dependency on human support teams.

Administrators interact with the system through a dedicated dashboard that provides tools for managing products, orders, customers, payments, and overall business performance. The dashboard presents key performance indicators through visual analytics, enabling business decisions to be driven by data rather than intuition. The admin panel also supports role-based access control, audit logs, and configuration options for AI components, ensuring that the system can be governed and refined over time as business requirements evolve.

This project, therefore, represents a convergence of classical web engineering principles with modern artificial intelligence practices, delivering a unified platform that is both technically sound and commercially relevant. By addressing the specific shortcomings observed in existing local e-commerce platforms and embracing international best practices, the E-Commerce Website with AI-Powered Features aims to contribute a practical, scalable, and intelligent solution to the growing digital economy of Pakistan and similar emerging markets.

1.1 Brief

The E-Commerce Website with AI-Powered Features is designed as a unified, web-based commerce platform that connects customers, sellers, and administrators through a single intelligent digital ecosystem. The aim of the project is to overcome the limitations of traditional online stores, where product discovery, search, and customer support are often disconnected and inefficient. The platform consolidates essential commerce services such as browsing, searching, purchasing, paying, and post-sale support into one integrated system and enhances them with AI components that deliver personalized experiences. The system leverages contemporary web technologies to provide a reliable, scalable, and user-friendly shopping experience that is accessible from any modern browser and adaptable to a wide range of devices.

1.1.1 Purpose of the Project

The primary purpose of the E-Commerce Website with AI-Powered Features is to centralize online retail operations by offering an all-in-one digital platform that supports product discovery, smart search, personalized recommendations, secure checkout, payment integration, order management, and AI-driven customer support. The project aims to simplify the interaction between customers and online businesses by removing the friction that typically exists in conventional online stores. It seeks to empower customers by giving them intelligent tools to find products quickly, make informed decisions through recommendations, and receive instant help through a chatbot. For administrators, the system provides a comprehensive control centre with analytics, management

tools, and extensible AI configurations, thereby enabling them to operate a modern, data-driven digital business.

1.1.2 Scope of the System

The scope of the E-Commerce Website with AI-Powered Features extends across multiple functional areas of online retail. On the customer side, the platform handles account registration, authentication, profile management, catalogue browsing, smart search, product recommendations, cart management, checkout, payment processing, order tracking, and customer support through an integrated chatbot. On the administrative side, it covers product and category management, order processing, customer management, seller approval, inventory tracking, financial reporting, and analytics. The AI components cover personal recommendations, NLP-enhanced search, and a conversational chatbot that can answer product and order queries. The system is designed to work as a full-fledged online store capable of supporting small and medium businesses and can be extended in the future to support mobile applications, multi-vendor marketplaces, and advanced analytics.

1.1.3 Technologies Used

The E-Commerce Website with AI-Powered Features is developed using the MERN technology stack, which combines MongoDB, Express.js, React.js, and Node.js into a cohesive full-stack JavaScript ecosystem. React.js is used for the frontend, providing modular and component-based architecture that supports rapid development of dynamic interfaces and ensures a consistent user experience across pages. Node.js serves as the runtime environment for the backend, offering non-blocking, event-driven capabilities that handle concurrent requests efficiently. Express.js is used on top of Node.js to define the REST API structure, middleware, and routing logic. MongoDB is selected as the primary database because of its flexibility in storing diverse product information, user profiles, reviews, and AI-related data structures in a schema-less, document-oriented format. Additional tools and libraries include JWT for authentication, bcrypt for password hashing, Stripe or similar gateway integrations for payments, and various machine-learning libraries for the AI modules.

1.1.4 System Overview

The E-Commerce Website with AI-Powered Features operates as an interconnected system that supports users at multiple levels. Customers can create accounts, maintain profiles, browse an organized catalogue, search for products using intelligent queries, receive recommendations based on their behaviors, add items to a shopping cart, complete secure payments, and track the status of their orders. Sellers and administrators interact with the platform through a dedicated dashboard that provides tools for managing the catalogue, monitoring orders, viewing performance metrics, and overseeing the overall operation of the store. Behind the scenes, the AI components continually analyze user behaviors, update recommendation models, and improve the accuracy of search results. All interactions between the front end, backend, and AI components are coordinated through a well-defined set of REST APIs, ensuring clean separation, scalability, and maintainability.

1.1.5 Outcome of the Project

The project results in a complete and intelligent e-commerce ecosystem that enhances the quality and efficiency of online retail. The integrated structure saves time for customers, reduces manual effort for sellers and administrators, prevents inconsistencies in product information, and increases engagement through intelligent features. The outcome is a secure, reliable, and extensible platform that can be used as a foundation for real-world deployment, further academic research, or commercial adaptation. The inclusion of AI capabilities elevates the platform beyond a conventional online store, positioning it as a forward-looking solution that aligns with international trends in digital commerce. Moreover, the project delivers a comprehensive set of artifacts including requirement specifications, design documents, source code, test cases, and deployment guidelines, all of which together form a rich educational resource and a practical product.

1.2 Relevance to Course Modules

The E-Commerce Website with AI-Powered Features draws upon a wide spectrum of knowledge and skills acquired during the Bachelor of Science in Computer Science programmed at COMSATS University Islamabad, Sahiwal Campus. The project is not the application of a single course module, but rather an integration of several key areas of computer science, each of which contributes concepts, tools, and methods to the final system. The following sub-sections describe how different course modules are directly reflected in the design and development of the project.

1.2.1 Software Engineering

The Software Engineering module provides the foundation for planning, managing, and delivering the project in a disciplined manner. Concepts such as requirement elicitation, feasibility analysis, use case modelling, system design, iterative development, testing, and quality assurance are directly applied throughout the lifecycle of the system. The E-Commerce Website with AI-Powered Features follows an Agile-inspired methodology that incorporates incremental delivery, regular review cycles, and continuous improvement, all of which are principles taught in the Software Engineering curriculum. The software design documentation, use case descriptions, and test plans included in this report are structured in accordance with the standards covered in the course.

1.2.2 Database Systems

The Database Systems module contributes directly to the design and implementation of the data layer of the system. Although the platform uses MongoDB, which is a NoSQL document-oriented database, the underlying principles of data modelling, normalization, relationships, indexing, and query optimization remain highly relevant. Concepts such as entity-relationship modelling, schema design, transactions, and data integrity are adapted to a document-oriented context, resulting in a set of well-designed collections for users, products, orders, categories, carts, reviews, and AI-related data. The module also provides the theoretical background for understanding how queries are executed and how performance can be improved through indexing and aggregation pipelines.

1.2.3 Web Engineering and Internet Programming

The Web Engineering and Internet Programming module is arguably the most directly relevant course, as the entire project is delivered as a web application. The principles of responsive design, client-server communication, HTTP protocols, REST APIs, and modern JavaScript frameworks are at the core of the system. React.js is used to implement a component-based frontend, while Node.js and Express.js are used to build backend services. The knowledge of HTML5, CSS3, JavaScript, and advanced tooling such as package managers, bundlers, and version control systems has been essential for the successful development of the platform.

1.2.4 Mobile Application Development

Although the current version of the project is delivered as a web application, the Mobile Application Development module has influenced design in important ways. The user interface is designed to be fully responsive, supporting a wide range of screen sizes from desktop monitors to tablets and smartphones. The system is also structured in a way that allows a mobile application, developed in React Native or a similar framework, to be added in a future iteration. Many of the concepts covered in the mobile course, such as touch-friendly interactions, performance optimization, and offline-aware design, are reflected in the choices made for the web interface.

1.2.5 Object Oriented Programming

Object Oriented Programming concepts such as encapsulation, abstraction, inheritance, and polymorphism are applied throughout the codebase, even though JavaScript is a prototype-based language rather than a classical object-oriented language. The frontend components, backend models, services, and controllers are organized around classes and objects that encapsulate state and behaviors. The separation of concerns between models, views, and controllers, as well as the modular structure of the React components, are directly influenced by object-oriented design principles.

1.2.6 Data Structures and Algorithms

The Data Structures and Algorithms module provides the essential toolkit for implementing the AI components and optimizing the performance of the system. Efficient data structures such as hash maps, trees, and graphs are used in the recommendation engine and search modules. Algorithms such as sorting, searching, and graph traversal are applied to rank products, compute similarities, and navigate catalogues. The analysis of time and space complexity ensures that the AI components and the core e-commerce workflows perform well even as the number of users and products grows.

1.2.7 Human Computer Interaction

The Human Computer Interaction module shapes the overall design of the user interface, ensuring that the platform is intuitive, accessible, and pleasant to use. Concepts such as user-centered design, usability heuristics, accessibility guidelines, information architecture, and visual hierarchy are applied throughout the design process. The layout of pages, the grouping of functionalities, the use of colors and typography, the feedback given for user actions, and the handling of errors all follow established HCI principles. The goal is to create an experience that is easy to learn for new users and efficient for returning users.

1.2.8 Information Security

Information Security plays a central role in the development of the E-Commerce Website with AI-Powered Features, given that the platform handles sensitive user data, payment information, and account credentials. The module provides theoretical background for concepts such as authentication, authorization, encryption, input validation, session management, and secure communication. These concepts are reflected in the implementation through JWT-based authentication, bcrypt-based password hashing, HTTPS for data in transit, parameterized queries, protection against common web vulnerabilities such as XSS and CSRF, and careful handling of payment data through trusted gateways.

1.2.9 Artificial Intelligence and Machine Learning

Artificial Intelligence and Machine Learning modules form the theoretical basis for the AI components of the system. Concepts such as supervised learning, collaborative filtering, content-based filtering, natural language processing, vector representations, and similarity measures are directly applied in the recommendation engine, the smart search, and the chatbot. The project demonstrates how these concepts can be translated from theoretical models into practical components that operate on real user data within the context of a production-like web application.

1.2.10 Final Year Project I and II

Final Year Project I and II provide the overall framework in which this project is developed. The two-semester structure allows the team to move from idea generation and requirement analysis in the first semester to detailed design, implementation, and testing in the second. Project planning, supervisor meetings, progress tracking, risk management, documentation, and presentation skills are all honed through these two courses. The final report itself is an outcome of these modules, and its structure follows the conventions expected of a final year project at the department level.

1.3 Project Background

The idea of developing an E-Commerce Website with AI-Powered Features emerged from observing the current state of online shopping in Pakistan and contrasting it with the best practices adopted by leading international platforms. While global giants such as Amazon, eBay, and Alibaba heavily invest in artificial intelligence to improve customer experience, local platforms often lag behind, offering only basic product listings and limited personalization. At the same time, the number of internet users and online shoppers in Pakistan has grown dramatically, creating both a need and an opportunity for a modern, intelligent, and locally relevant e-commerce platform.

The project is also motivated by the observation that many small and medium businesses in Pakistan struggle to establish a strong online presence. They often rely on social media pages, WhatsApp messages, and manual order-taking to sell their products. Such approaches, while accessible, lack structure, scalability, and the ability to provide a consistent shopping experience to customers. By offering a modern, AI-powered platform that can be adapted to different categories of products, the project aims to provide a technological foundation that such businesses can adopt and customize to meet their needs.

Additionally, the rapid evolution of web and AI technologies has made it possible for student teams to build systems of considerable complexity as part of their final year project. Modern open-source

tools, cloud services, and educational resources have lowered the barrier to entry for building full-stack applications with integrated machine learning components. This creates an ideal environment for a project that not only fulfils academic requirements but also demonstrates practical relevance and potential for real-world deployment.

The project is inspired by three main categories of existing systems. The first category includes mainstream e-commerce platforms that provide a rich, but often generic, shopping experience. The second category includes recommendation-focused research projects that have shown the effectiveness of collaborative and content-based filtering in increasing user engagement and sales. The third category includes chatbot-based customer service solutions that have transformed the way support is delivered into digital businesses. The E-Commerce Website with AI-Powered Features combines insights from these three categories into a unified, practical, and locally relevant platform.

1.4 Literature Review

A review of existing literature and industry practice reveals that e-commerce platforms have gone through several waves of evolution, each accompanied by new technologies and new challenges. Early online stores focused primarily on static product catalogues and simple shopping cart systems. As the number of users and products increased, search and navigation became important research areas, leading to the development of sophisticated indexing and ranking techniques. Recommendation systems subsequently emerged as a critical component, with collaborative filtering and content-based approaches being extensively studied and deployed. More recently, the rise of deep learning and natural language processing has enabled more advanced forms of personalization, including intent recognition, semantic search, conversational commerce, and visual search.

Several studies have explored the impact of AI on e-commerce. Research in recommender systems has shown that personalized recommendations can significantly increase conversion rates, average order value, and customer loyalty. Studies on chatbot-based customer support have demonstrated that well-designed conversational agents can handle a substantial portion of routine queries, freeing human agents to focus on more complex issues. NLP-driven search has been shown to reduce the gap between user intent and product discovery, especially for users who are not familiar with specific product terminology.

Local literature and observations highlight that many Pakistani e-commerce platforms have yet to fully adopt these advanced capabilities. While there has been progress in areas such as payment integration and delivery logistics, the intelligence layer of most local platforms remains limited. This gap is particularly visible in small and medium-sized online businesses that lack the technical resources to build and maintain sophisticated AI systems. Academic projects such as this one can play an important role in demonstrating how accessible and effective AI-powered e-commerce can be when built using freely available tools and techniques.

In the context of this project, the literature review has also considered the architectural patterns commonly used for full-stack JavaScript applications. Patterns such as the MVC architecture, service-oriented design, and component-based frontends are widely accepted as effective approaches to building scalable and maintainable web applications. The literature suggests that the MERN stack is particularly well-suited for rapid development of modern e-commerce platforms

due to the alignment of its components with contemporary best practices and the large ecosystem of open-source libraries surrounding it.

Table 1: Comparison of Existing E-Commerce Applications and the Proposed System

Feature	Typical Local Platforms	International Platforms	Proposed System
Catalogue Management	Basic	Advanced	Advanced with AI metadata
Search	Keyword-based	Semantics + Personalized	AI-powered smart search
Recommendations	Limited / Absent	Extensive	Personalized AI recommendations
Chatbot Support	Rare	Common	Integrated AI chatbot
Payment Options	Limited	Broad	Multiple gateways supported
Analytics	Basic	Advanced	Dashboard with KPIs
Scalability	Moderate	High	High (cloud-ready)
Security	Variable	Strong	JWT + HTTPS + Best Practices

1.5 Analysis from Literature Review

The analysis of the literature review provides valuable guidance for designing the E-Commerce Website with AI-Powered Features. It identifies the gaps in existing systems, confirms the relevance of AI-based features, and highlights the architectural patterns that should be adopted to build a robust, scalable, and user-friendly platform. The following sub-sections describe how specific insights from the literature are translated into design decisions for this project.

1.5.1 Alignment with Digital Commerce Adoption Trends

The literature clearly indicates that digital commerce is growing rapidly across both developed and emerging markets, driven by increased internet penetration, improved digital payment infrastructure, and changing consumer behaviors. The E-Commerce Website with AI-Powered Features is aligned with these trends by offering a modern online shopping experience that takes

advantage of web and AI technologies. The platform is designed to serve a broad range of users, from tech-savvy customers who appreciate advanced features to first-time online shoppers who require simple, intuitive navigation.

1.5.2 Comparison with Cloud Based E-Commerce Research

Cloud-based architecture is widely recognized in literature as a powerful approach for building scalable, reliable, and cost-efficient e-commerce platforms. The project adopts cloud-ready design principles by separating the frontend, backend, and database, and by using technologies that can be deployed on a variety of cloud platforms such as Vercel, AWS, Azure, and Digital Ocean. MongoDB can be hosted on cloud services such as MongoDB Atlas, providing geographic distribution, automated backups, and high availability.

1.5.3 Relevance to Existing Recommendation and Search Systems

Research on recommendation systems provides the theoretical basis for the AI-powered recommendation engine of this project. Both collaborative filtering and content-based filtering are considered, and a hybrid approach is adopted to balance the strengths of each method. Similarly, research on intelligent search systems informs the design of the smart search module, which uses techniques such as tokenization, stemming, stop-word removal, synonym expansion, and vector similarity to improve the relevance of search results.

1.5.4 Integration of AI-Powered Features

The literature highlights the growing importance of AI in transforming customer experience in e-commerce. The E-Commerce Website with AI-Powered Features integrates several AI-driven capabilities, including personalized recommendations, intelligent search, and a conversational chatbot. Each of these capabilities is designed to address a specific pain point in the customer journey and to contribute to an overall more satisfying experience. The design ensures that the AI components can be updated, retrained, or replaced without disrupting the rest of the system.

1.5.5 Application of Security and Privacy Standards

Security and privacy are consistently highlighted in literature as critical concerns for e-commerce platforms, especially those that handle payment and personal information. The project applies widely recognized security practices, including HTTPS, secure authentication with JWT, hashed password storage, input validation, and protection against common web vulnerabilities. The system is also designed to minimize the exposure of sensitive data by isolating payment processing through trusted gateways.

1.5.6 Involvement of Mobile Commerce Concepts

Mobile commerce is identified in the literature as a major growth area, with a significant share of online transactions now occurring on smartphones and tablets. The project addresses this trend by implementing a responsive web design that adapts gracefully to different screen sizes. Although a native mobile application is not included in the current scope, the system is designed to support future expansion to mobile platforms.

1.5.7 Addressing Limitations of Existing E-Commerce Technologies

The literature review identifies several limitations of existing e-commerce technologies, including poor personalization, limited search capabilities, inadequate customer support, and fragmented user experiences. The E-Commerce Website with AI-Powered Features addresses these limitations directly by providing a unified platform that integrates AI-driven recommendations, intelligent search, and conversational support within a single, coherent system. The design also emphasizes maintainability and extensibility, so that additional capabilities can be added as new research and technologies emerge.

1.6 Methodology and Software Lifecycle

The methodology adopted for the development of the E-Commerce Website with AI-Powered Features is based on Agile principles, which provide a flexible, iterative, and collaborative approach to software development. The Agile methodology is particularly well-suited to projects that combine established technologies, such as the MERN stack, with emerging technologies, such as AI components, because it allows requirements, designs, and implementations to be refined progressively as the team's understanding of the problem deepens.

1.6.1 Methodology

Agile methodology emphasizes continuous delivery, close collaboration with stakeholders, and the ability to respond to change. The project team organizes its work into sprints, each with clearly defined goals, deliverables, and acceptance criteria. At the end of each sprint, the team reviews the progress, collects feedback, and adjusts the plan accordingly. This approach allows the project to evolve in response to lessons learned during implementation and feedback received from the supervisor and end users.

1.6.2 Agile Practices Applied

Several Agile practices are applied throughout the lifecycle of the project. Regular stand-up meetings are held to discuss progress and address obstacles. A product backlog is maintained to capture and prioritize requirements. Sprint planning sessions are conducted to decide the scope of each iteration, and retrospectives are held at the end of each sprint to identify opportunities for improvement. Continuous integration practices are applied to ensure that changes to the codebase are integrated and tested frequently, reducing the risk of late-stage integration problems.

1.6.3 Software Development Life Cycle (SDLC) Model

The Software Development Life Cycle followed by the project is an iterative and incremental model that combines elements of Scrum with the classical phases of requirement analysis, design, implementation, testing, and deployment. Each iteration takes a subset of requirements from the backlog, moves them through design and implementation, and delivers a working increment of the system. This incremental approach allows early versions of the system to be tested and improved, ensuring that the final product is well-tuned and robust.

1.6.4 Product Backlog Management

The product backlog serves as the single source of truth for the requirements of the project. It contains user stories, technical tasks, and non-functional requirements that are continuously

refined and reprioritized. Each item in the backlog is described with sufficient detail to allow estimation and implementation. High-priority items are refined further and selected for inclusion in the next sprint, while lower-priority items remain in the backlog until they are needed.

1.6.5 Incremental Delivery

Incremental delivery is one of the core principles of the Agile methodology. Instead of delivering the entire system at the end of the project, the team delivers working increments at the end of each sprint. These increments are tested, reviewed, and demonstrated to the supervisor. This approach reduces risk, provides early visibility into the progress of the project, and ensures that the final system is a natural evolution of several well-tested intermediate versions.

1.6.6 Rationale behind Selected Methodology

The selection of Agile methodology is motivated by the nature of the project, which combines well-understood e-commerce workflows with emerging AI components. The former can be planned and developed in a relatively predictable manner, while the latter benefit from experimentation and iteration. Agile accommodates both types of work, allowing the team to apply disciplined engineering to the core platform while exploring and refining the AI components through short cycles of experimentation and integration.

1.6.7 Development Tools and Environment

The development environment consists of modern tools that support productivity and collaboration. Visual Studio Code is used as the primary integrated development environment, with a rich set of extensions for JavaScript, TypeScript, React, and Node.js development. Git and GitHub are used for version control and collaboration. Postman and Insomnia are used for testing REST APIs. MongoDB Compass is used as a graphical interface for inspecting and managing the database. The project is configured to run locally using Node.js and can be deployed to cloud platforms such as Vercel, Render, or AWS.

1.6.8 Testing and Quality Assurance

Testing and quality assurance are integrated into every phase of the development lifecycle. Unit tests verify the behaviors of individual components, functional tests verify specific features, integration tests verify the interaction between components, and system tests verify the behaviors of the platform as a whole. Automated test suites are used where possible to reduce the cost of repeated testing, while manual testing is used for visual, exploration, and user experience checks. Quality assurance activities also include code reviews, static analysis, and performance profiling.

1.6.9 Advantages of the Selected Methodology and SDLC

The selected methodology and SDLC bring several advantages to the project. They allow the team to start delivering value early and continuously, reduce the risk of misunderstanding the requirements, improve the quality of the product through frequent testing, and provide a natural framework for integrating AI components alongside traditional e-commerce features. They also prepare the team for professional software development practices that are widely used in the industry, making the project an effective bridge between academic study and future careers.

Chapter 2

PROBLEM DEFINITION

2. Problem Definition

The problem definition phase is a critical step in the development of any software system. It provides a precise articulation of the issues that the system is intended to resolve, identifies the delivery and development requirements, and examines the limitations of the current situation. This chapter presents the problem statement of the E-Commerce Website with AI-Powered Features, outlines the expected deliverables, describes the development requirements, and analyses the shortcomings of existing e-commerce practices that motivate the need for a more modern, intelligent, and unified platform.

2.1 Problem Statement

Despite the rapid growth of online shopping in Pakistan and similar emerging markets, the quality of digital commerce experiences remains inconsistent and, in many cases, substantially below international standards. Customers visiting local e-commerce platforms often face difficulties in finding the products that best match their needs, are exposed to irrelevant suggestions, and struggle to obtain timely support when problems arise. Sellers and store administrators, on the other hand, manage their operations using fragmented tools that do not provide a clear, unified view of their business. The result is a sub-optimal shopping experience, reduced customer satisfaction, and constrained business growth.

The primary problem addressed by the E-Commerce Website with AI-Powered Features is the absence of an integrated, intelligent, and user-centric online shopping platform that serves both customers and administrators with modern capabilities. More specifically, the system seeks to resolve the following interconnected problems: product discovery limitations due to simplistic search; the lack of personalized recommendations that align with individual user behaviors; inadequate customer support caused by reliance on manual communication channels; insufficient security measures in many local platforms; the absence of comprehensive analytics for administrators; and the fragmentation of different digital commerce functions across multiple disconnected tools. By addressing these issues within a single platform, the project aims to deliver a meaningful improvement over the current state of online shopping.

2.2 Deliverables and Development Requirements

This section describes the key deliverables expected from the project and the development requirements that must be satisfied to realize them. The deliverables represent the tangible outputs of the project, while the development requirements describe the technical and non-technical conditions under which these outputs will be produced.

2.2.1 Deliverables

The project produces a set of deliverables that together constitute a comprehensive digital commerce platform. These deliverables are described in the following sub-sections.

2.2.1.1 Customer Web Application

The customer web application is the primary interface through which end users interact with the platform. It supports registration and authentication, profile management, product browsing, smart search, personalized recommendations, cart management, checkout and payment, order tracking,

product reviews, and interaction with the AI-powered chatbot. The interface is designed to be intuitive, responsive, and accessible from a wide range of devices, ensuring consistent experience for both desktop and mobile users.

2.2.1.2 Admin Dashboard

The admin dashboard is a dedicated interface for administrators and authorized staff. It provides tools for managing products, categories, inventory, orders, customers, sellers, and payments. The dashboard also includes analytics views that present key performance indicators such as sales, revenue, top products, and customer behaviors patterns. Role-based access control ensures that different administrators can perform different tasks according to their responsibilities.

2.2.1.3 AI Recommendation Engine

The AI recommendation engine is a core deliverable that differentiates this project from traditional e-commerce platforms. It analyses user behaviors, product attributes, and historical interactions to generate personalized recommendations. The engine supports multiple recommendation strategies, including collaborative filtering, content-based filtering, and hybrid approaches. It exposes its functionality through REST APIs that can be consumed by the frontend.

2.2.1.4 Intelligent Chatbot

The intelligent chatbot is an integrated conversational agent that provides immediate assistance to customers. It can answer questions about products, orders, delivery, returns, and account management, and can intelligently hand over complex queries to human support agents. The chatbot is accessible from most pages of the platform and is designed to be friendly, helpful, and consistent with the overall brand of the store.

2.2.1.5 Centralized Cloud Database

The centralized cloud database is implemented using MongoDB and stores all the structured and semi-structured data required by the platform, including users, products, orders, reviews, cart sessions, chat logs, and AI-related data structures. The database is designed to support high availability, scalability, and secure access. It can be deployed on services such as MongoDB Atlas for production environments.

2.2.2 Development Requirements

In addition to the deliverables, the project has several development requirements that define the technical and organizational conditions under which the system is built.

2.2.2.1 Technical Requirements

The technical requirements include the use of the MERN stack, adherence to REST API design principles, responsive web design, secure authentication with JWT, password hashing with bcrypt, integration with trusted payment gateways, and modular architecture for AI components. Source code must be managed in a version control system, development must follow a coding standard, and automated tests must be used where practical. The system must be deployable to both local and cloud environments, and it must be documented sufficiently to allow future developers to understand and extend it.

2.3 Current System

Before designing a new system, it is important to understand the limitations of the current situation. This section presents an analysis of how online shopping is typically conducted in Pakistan today and highlights the problems that motivate the need for a more intelligent and integrated platform.

2.3.1 Manual Shopping and Record-Keeping

A significant portion of retail in Pakistan is still conducted through physical stores, phone orders, and informal digital channels such as WhatsApp and social media pages. Even when online stores exist, many of them rely heavily on manual record-keeping, with sellers tracking orders in spreadsheets, paper notebooks, or simple ledgers. This manual approach is prone to errors, difficult to scale, and limits the ability of businesses to analyze their operations or respond quickly to changes in demand. The absence of a unified digital record makes it challenging to track customer history, inventory movement, and financial performance.

2.3.2 Fragmented Online Storefronts

Many small and medium businesses operate fragmented online storefronts spread across multiple platforms. A single business might have a Facebook page, an Instagram account, a WhatsApp catalogue, and perhaps a basic website or a listing on a general marketplace. Each of these channels has its own way of displaying products, taking orders, and communicating with customers. This fragmentation leads to inconsistent information, customer confusion, and significant operational overhead for the seller. It also makes it almost impossible to implement advanced features such as personal recommendations or unified analytics.

2.3.3 Limited Accessibility and Communication

In many existing online stores, customer communication is limited to email or phone, and sometimes to informal channels such as chat messages. Response times can be slow, and customers may have to repeat the same information to different agents. The lack of structured communication tools prevents businesses from keeping complete records of customer interactions and from gaining insights into common concerns and feedback. Accessibility is further limited by the absence of features such as multilingual support, responsive design, and accessibility compliance.

2.3.4 Security and Privacy Concerns

Security and privacy are significant concerns in many local e-commerce platforms. Some platforms store passwords in plaintext, fail to use HTTPS consistently, or expose sensitive APIs without proper authentication. Payment processing is sometimes handled by unverified intermediaries, raising the risk of fraud and data leaks. These weaknesses not only endanger customers but also erode trust in online shopping, making it harder for legitimate businesses to grow their customer base.

2.3.5 Inadequate Integration of Intelligent Tools

Most local e-commerce platforms offer little or no integration of intelligent tools. Search functionality is typically restricted to simple keyword matching, with no support for intent recognition, synonyms, or personalization. Recommendation systems, when present, are limited to rudimentary logic such as "most popular" or "recently added". Chatbots are rare, and when

deployed, are often rigid scripted agents rather than intelligent conversational systems. These limitations translate directly into lower engagement, lower conversion rates, and a less satisfying shopping experience for customers.

2.3.6 Regional Disparities

There are significant regional disparities in access to quality e-commerce services across Pakistan. Customers in major cities such as Karachi, Lahore, and Islamabad benefit from better internet connectivity, wider delivery coverage, and more diverse product catalogues. Customers in smaller cities and rural areas often face higher shipping costs, longer delivery times, and limited access to international products. A well-designed cloud-based platform can help reduce these disparities by offering consistent experience regardless of location and by supporting logistics partners that can reach a wider geographic area.

2.3.7 Summary of Limitations

In summary, the current state of e-commerce in Pakistan is characterized by a mixture of traditional methods, fragmented online storefronts, limited intelligent functionality, security concerns, and regional disparities. While significant progress has been made in some areas, there remains a clear opportunity to improve the quality of online shopping through a unified, intelligent, and secure platform. The E-Commerce Website with AI-Powered Features is designed to address these issues through a careful combination of modern web technologies, AI capabilities, and user-centered design.

2.4 Summary

This chapter has defined the core problem addressed by the project, identified the deliverables and development requirements, and analyzed the limitations of the current system. The analysis confirms that there is a genuine need for a modern, intelligent, and unified e-commerce platform that can serve both customers and administrators effectively. It also provides a clear justification for the choice of technologies and AI components adopted by the project. The next chapter builds on this foundation by presenting a detailed requirement analysis, including use cases, functional requirements, and non-functional requirements.

Chapter 3

REQUIREMENT ANALYSIS

3. Requirement Analysis

3.1 Introduction to Requirement Analysis

Requirement analysis is the process of identifying, specifying, and organizing the needs of all stakeholders of a software system. It forms the foundation upon which the rest of the project is built, and its quality has a direct impact on the success of the final product. The E-Commerce Website with AI-Powered Features serves multiple categories of users, each with distinct needs and expectations. Accordingly, the requirement analysis for this project is performed systematically, starting with an overview of the system, identifying its actors, describing the use cases, and formulating both functional and non-functional requirements in detail.

The goal of this chapter is to present a clear, comprehensive, and unambiguous description of what the system must do and how it must behave. The requirements are derived from a combination of literature review, analysis of existing systems, discussions with the supervisor, and observations of typical customer journeys online. They are structured in a way that supports traceability, allowing each feature in the implementation to be traced back to a specific requirement.

3.2 System Overview

The E-Commerce Website with AI-Powered Features is a web-based platform that allows customers to discover, evaluate, and purchase products through a user-friendly interface enhanced by AI capabilities. The system supports multiple user roles, including customers, sellers, administrators, and guest users. It provides a unified experience that combines product browsing, smart search, personalized recommendations, shopping cart management, secure checkout, order tracking, and customer support. The AI components are integrated into the platform through REST APIs that allow the frontend to consume intelligent features in a consistent manner.

At a high level, the system consists of three main layers. The first layer is the presentation layer, implemented as a React-based single-page application that handles all user interactions. The second layer is the application layer, implemented using Node.js and Express.js, which exposes REST APIs and contains the business logic of the platform. The third layer is the data layer, implemented using MongoDB, which stores persistent data such as users, products, orders, reviews, and AI models. A set of supporting services, such as payment gateways and external notification systems, are integrated through additional APIs.

3.3 Actors in the System

The system defines several actors, each representing a category of users with specific responsibilities and privileges. The following sub-sections describe the main actors and their roles.

3.3.1 Customer

The customer is the primary actor of the system. Customers create accounts, sign in, browse products, search using intelligent queries, view recommendations, add items to a shopping cart, place orders, make payments, track deliveries, write reviews, and interact with the chatbot. Customers are also able to manage their profiles, update personal information, view order history, and save favorite products.

3.3.2 Seller / Vendor

In future extensions of the platform, the seller or vendor actor will represent third-party businesses that list their own products on the platform. Sellers can register, submit their business credentials for verification, add and manage their own products, view orders placed for their items, and monitor their performance. In the current scope, this actor is primarily supported through the admin panel, but the design ensures that the full multi-vendor capability can be added in the future.

3.3.3 Administrator

The administrator is responsible for managing the overall platform. Administrators approve sellers, manage the product catalogue, monitor orders, oversee customer accounts, configure AI components, view analytics, and resolve escalated customer issues. Different administrators may be granted different levels of access through role-based access control, allowing the platform to support a diverse administrative team.

3.3.4 Guest User

The guest user represents visitors who have not yet created an account. Guest users can browse the catalogue, perform searches, and view product details, but they must register or sign in before adding items to a cart, placing an order, or writing a review. The system is designed to encourage guest users to become registered customers by highlighting the benefits of registration, including personalized recommendations and order tracking.

3.4 Use Case Diagrams

The case diagram of the system represents the relationships between the actors and the main functionalities of the platform. It provides a high-level visual summary of what the system offers to each category of user. The diagram shows the customer interacting with use cases such as Login, Browse Products, Smart Search, Add to Cart, Checkout, Track Order, View Recommendations, and Chatbot Interaction. The administrator interacts with use cases such as Manage Products, Manage Orders, Manage Users, Approve Sellers, and View Analytics. The seller interacts with use cases such as Manage Own Products and View Own Orders, while the guest user interacts with Browse Products and Smart Search, with clear prompts to register for additional functionality.

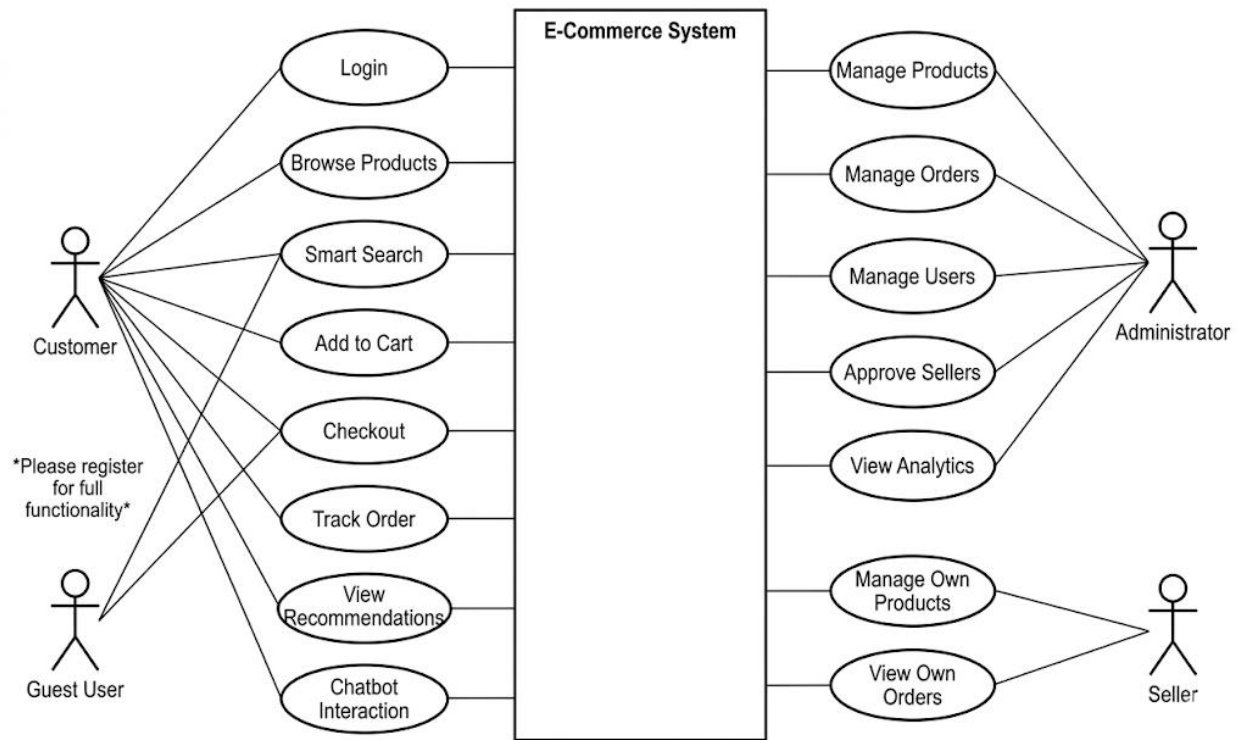


Figure 1 use Case diagram of E-Commerce system

3.5 Use Case Descriptions

This section provides structured descriptions for the main use cases of the system. Each use case is described using a consistent template that includes the use case identifier, name, actors, preconditions, main flow, alternative flow, postconditions, and exceptions.

3.5.1 Login/Register Use Case

Table 2: Structured Description of the Login/Register Use Case

Use Case	Description
Use Case ID	UC-01
Name	Login / Register
Actors	Customer, Seller, Administrator, Guest User
Preconditions	The user has access to the platform and a working internet connection.

Main Flow	1. The user opens the login or registration page. 2. For registration, the user enters name, email, password, and optional profile information. 3. The system validates the input and creates a new account in the database. 4. For login, the user enters email and password. 5. The system verifies the credentials and issues a JWT token. 6. The user is redirected to the home page or the requested page.
Alternative Flow	If the user forgets the password, they can request a reset email containing a secure link to set a new password.
Postconditions	The user is authenticated and has access to features appropriate to their role.
Exceptions	Invalid credentials, duplicate registration, expired reset token, network failure.

3.5.2 Set Profile Use Case

Table 3: Structured Description of the Set Profile Use Case

Use Case	Description
Use Case ID	UC-02
Name	Set Profile
Actors	Customer
Preconditions	The user is logged into the system.
Main Flow	1. The user opens the profile page. 2. The user updates personal information such as name, email, phone, shipping addresses, and preferences. 3. The system validates the input and updates the database. 4. A confirmation message is displayed.
Alternative Flow	The user uploads a profile picture, which is stored and linked to the account.
Postconditions	Profile information is updated and immediately reflected across the system.

Exceptions	Invalid data, upload failure, unauthorized modification attempts.
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3.5.3 Browse Products Use Case

Table 4:Browse Products Use Case

Use Case	Description
Use Case ID	UC-03
Name	Browse Products
Actors	Customer, Guest User
Preconditions	The product catalogue is available and the platform is reachable.
Main Flow	1. The user opens the home page. 2. The user selects a category or scrolls through featured products. 3. The system displays the relevant products with pagination and sorting options. 4. The user filters results by price, brand, rating, or attributes.
Alternative Flow	The user clicks on a promotional banner, which opens a curated collection of products.
Postconditions	The user views a list of products that match the selected criteria.
Exceptions	Network failure, empty category, invalid filter combination.

3.5.4 Smart Search Use Case

Table 5:Smart Search Use Case

Use Case	Description
Use Case ID	UC-04
Name	Smart Search
Actors	Customer, Guest User

Preconditions	The search service is operational and the catalogue is indexed.
Main Flow	1. The user enters a query in the search bar. 2. The system analyses the query using NLP techniques. 3. Results are ranked using similarity, popularity, and personalization signals. 4. The user views ranked results and can refine the query.
Alternative Flow	The system suggests spelling corrections and related terms if no exact matches are found.
Postconditions	The user views a personalized and ranked list of search results.
Exceptions	Empty result, service unavailable, index corruption.

3.5.5 Add to Cart Use Case

Table 6:Add to Cart Use Case

Use Case	Description
Use Case ID	UC-05
Name	Add to Cart
Actors	Customer
Preconditions	The user is authenticated and the product is available in stock.
Main Flow	1. The user selects a product. 2. The user chooses quantity and options (size, color, etc.). 3. The user clicks Add to Cart. 4. The system updates the cart and displays a confirmation.
Alternative Flow	The user updates quantity or removes items from the cart at any time.
Postconditions	The cart reflects the updated selection.
Exceptions	Out-of-stock items, invalid quantity, cart limit exceeded.

3.5.6 Checkout and Payment Use Case

Table 7:Checkout and Payment Use Case

Use Case	Description
Use Case ID	UC-06
Name	Checkout and Payment
Actors	Customer
Preconditions	The cart contains at least one product and the user is authenticated.
Main Flow	1. The user opens the cart and proceeds to check out. 2. The user selects or enters a shipping address. 3. The user selects a payment method (card, wallet, cash on delivery). 4. The system validates the selection and initiates the payment through the payment gateway. 5. On success, an order is created and confirmation is displayed.
Alternative Flow	The user applies a coupon code or uses loyalty points before completing the payment.
Postconditions	A new order is created, inventory is updated, and the customer receives a confirmation.
Exceptions	Payment declined, invalid address, gateway timeout, inventory change during checkout.

3.5.7 View Recommendations Use Case

Table 8:View Recommendations Use Case

Use Case	Description
Use Case ID	UC-07
Name	View Recommendations
Actors	Customer

Preconditions	The user is authenticated and has some interaction history, or the system has enough anonymous signals.
Main Flow	1. The user opens the home page or a product detail page. 2. The system requests recommendations from the AI engine. 3. Recommendations are displayed in sections such as "Recommended for you" and "Customers also viewed".
Alternative Flow	If no personal history is available, the system falls back to popular products.
Postconditions	The user is presented with personalized suggestions.
Exceptions	Recommendation service unavailable, empty result, timeout.

3.5.8 Chatbot Interaction Use Case

Table 9: Chatbot Interaction Use Case

Use Case	Description
Use Case ID	UC-08
Name	Chatbot Interaction
Actors	Customer, Guest User
Preconditions	The chatbot service is operational.
Main Flow	1. The user opens the chat window. 2. The user enters a question into natural language. 3. The chatbot interprets the question and returns an answer or suggested actions. 4. The user may continue the conversation.
Alternative Flow	If the chatbot cannot answer, the conversation is escalated to a human agent.
Postconditions	The user receives assistance and the conversation is logged for future improvement.
Exceptions	Service failure, misunderstanding of query, unsupported language.

3.5.9 Track Order Use Case

Table 2. 1 Track Order Use Case

Use Case	Description
Use Case ID	UC-09
Name	Track Order
Actors	Customer
Preconditions	The user has at least one active order.
Main Flow	1. The user opens the My Orders section. 2. The system displays a list of current and past orders. 3. The user selects an order to view details, including status, estimated delivery, and tracking information.
Alternative Flow	The user cancels an order that has not yet been shipped, subject to cancellation policies.
Postconditions	The user has up-to-date visibility of the status of their orders.
Exceptions	Order not found, cancellation not permitted, courier service unavailable.

3.5.10 Review and Rate Product Use Case

Table 10:Review and Rate Product Use Case

Use Case	Description
Use Case ID	UC-10
Name	Review and Rate Product
Actors	Customer
Preconditions	The user has purchased the product and received it.
Main Flow	1. The user opens the product from the order history. 2. The user submits a star rating and a text review.

	3. The system validates and stores the review. 4. The review becomes visible on the product detail page after moderation
Alternative Flow	The user edits or deletes their own review within an allowed time window.
Postconditions	The product page reflects the updated rating and review.
Exceptions	Abusive content, duplicate review, unauthorized access.

3.5.11 Manage Products (Admin) Use Case

Table 11: Manage Products (Admin) Use Case

Use Case	Description
Use Case ID	UC-11
Name	Manage Products
Actors	Administrator
Preconditions	The administrator is authenticated with sufficient privileges.
Main Flow	1. The administrator opens the product management section. 2. The administrator creates, updates, or removes products and categories. 3. The administrator uploads product images and sets prices, stock, and attributes. 4. The system validates input and updates the catalogue.
Alternative Flow	The administrator performs bulk imports from a structured file.
Postconditions	The product catalogue is updated and immediately available to customers.
Exceptions	Invalid image format, duplicate SKU, insufficient privileges.

3.5.12 Manage Orders (Admin) Use Case

Table 12:Manage Orders (Admin) Use Case

Use Case	Description
Use Case ID	UC-12
Name	Manage Orders
Actors	Administrator
Preconditions	The administrator is authenticated and orders exist in the system.
Main Flow	1. The administrator opens the orders section. 2. The administrator filters orders by status, date, or customer. 3. The administrator updates order status, processes refunds, or issues shipments.
Alternative Flow	The administrator exports order reports for accounting purposes.
Postconditions	Order status is updated and relevant notifications are sent to customers.
Exceptions	Payment gateway failure, invalid status transition, stock discrepancy.

3.5.13 Manage Users (Admin) Use Case

Table 13:Manage Users (Admin) Use Case

Use Case	Description
Use Case ID	UC-13
Name	Manage Users
Actors	Administrator
Preconditions	The administrator is authenticated with user management privileges.
Main Flow	1. The administrator opens the user management section. 2. The administrator searches for a user or filters by role.

	3. The administrator views, updates, suspends, or deletes user accounts.
Alternative Flow	The administrator assigns or revokes roles and permissions.
Postconditions	User accounts are updated according to the changes.
Exceptions	Attempted removal of the last administrator, conflicts with active orders.

3.5.14 View Analytics (Admin) Use Case

Table 14:View Analytics (Admin) Use Case

Use Case	Description
Use Case ID	UC-14
Name	View Analytics
Actors	Administrator
Preconditions	The administrator is authenticated.
Main Flow	1. The administrator opens the analytics section. 2. The system displays charts for sales, revenue, conversion rate, top products, and user activity. 3. The administrator filters by time range or segment.
Alternative Flow	The administrator exports data for external analysis.
Postconditions	The administrator obtains actionable insights into the business.
Exceptions	Missing data, report generation timeout.

3.5.15 Approve Seller Registration Use Case

Table 15: Approve Seller Registration Use Case

Use Case	Description
Use Case ID	UC-15
Name	Approve Seller Registration
Actors	Administrator
Preconditions	A seller has submitted a registration request with supporting documents.
Main Flow	1. The administrator opens the pending seller approvals section. 2. The administrator reviews the submitted documents and verifies identity. 3. The administrator approves or rejects the application. 4. The system notifies the seller of the decision.
Alternative Flow	The administrator requests additional information from the seller before deciding.
Postconditions	The seller is either onboarding into the platform or rejected.
Exceptions	Incomplete documents, duplicate registration, policy violations.

3.6 Functional Requirements

Functional requirements describe the services that the system provides to its users. They represent the behaviors and features that the system must implement to meet the goals of the project. This section organizes the functional requirements into categories that align with the main modules of the system.

3.6.1 User Management

The system shall allow users to register, log in, log out, and reset their passwords. It shall support the management of user profiles, including personal information, addresses, and preferences. The system should enforce role-based access control, distinguishing between customers, sellers, administrators, and guest users. It shall maintain an audit trail of significant account actions such as password changes and role modifications.

3.6.2 Product Catalogue Management

The system shall allow administrators to create, update, and remove products and categories. It shall support the upload of multiple product images, the specification of variants such as size and color, and the configuration of pricing and inventory. The system shall allow customers to browse and filter products by category, brand, price, rating, and other attributes. It should also support featured products and promotional banners on the home page.

3.6.3 Shopping Cart and Checkout Management

The system should provide a persistent shopping cart that retains items across sessions for logged-in users. It should allow customers to update quantities, remove items, and apply coupon codes before checkout. The checkout process shall collect shipping and billing information, display the total cost including taxes and shipping, and confirm the order before processing payment. The system shall integrate with one or more payment gateways to process secure transactions.

3.6.4 Payment and Financial Management

The system shall support multiple payment methods, including credit/debit cards, digital wallets, and cash on delivery, subject to availability in the region. It shall record all financial transactions, generate invoices, and support refunds and partial refunds. Administrators shall be able to view financial reports, including revenue, refunds, and tax calculations.

3.6.5 AI Recommendation and Search

The system shall provide personalized product recommendations based on the user's browsing history, purchase history, and product similarity. It should support multiple recommendation strategies and allow administrators to configure which strategies are active. The smart search module shall interpret natural language queries, correct spelling errors, detect synonyms, and rank results using relevance and personalization signals. The AI components shall expose their functionality through REST APIs.

3.6.6 Chatbot and Customer Support

The system shall include a chatbot that can answer common questions about products, orders, delivery, returns, and accounts. The chatbot shall support escalation to human agents for complex queries. The system shall maintain a lot of conversations for training, improvement, and auditing purposes. The chatbot shall be accessible from most pages of the platform.

3.6.7 Admin and System Management

The system shall provide an admin dashboard for managing products, orders, customers, sellers, and AI components. It shall support role-based access control, audit logs, and configuration tools. Administrators shall be able to monitor system health, view usage statistics, and adjust platform settings through the dashboard.

3.6.8 Data Reporting and Analytics

The system should provide analytics views that display key performance indicators such as sales, revenue, conversion rate, top products, and user activity. It shall support filtering by time range,

category, and region. The system shall allow administrators to export reports in common formats such as CSV and PDF for use in external tools.

3.7 Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints under which the system must operate. They are essential to the success of the platform, as they determine how well the system performs under real-world conditions.

3.7.1 Performance

The system should provide fast response times for all major user interactions. Pages shall load within a reasonable time frame under normal load conditions, and the search and recommendation APIs shall return results within a few hundred milliseconds for typical queries. The system should be designed to handle concurrent users without significant degradation in performance.

3.7.2 Scalability

The system shall be scalable to support a growing user base, catalogue, and transaction volume. The architecture shall separate the frontend, backend, and database, allowing each component to be scaled independently. The AI components should be designed to support batch training and real-time inference, with scalable deployment options.

3.7.3 Security

The system should enforce strong security measures throughout. All communications shall use HTTPS. Passwords shall be stored using a strong hashing algorithm such as bcrypt. Authentication shall be performed using JWT tokens with appropriate expiry and refresh mechanisms. The system shall protect against common web vulnerabilities including cross-site scripting, cross-site request forgery, SQL injection, and insecure direct object references. Payment processing shall be delegated to trusted gateways to minimize the exposure of sensitive data.

3.7.4 Reliability and Availability

The system should be reliable and highly available. It shall include error handling, graceful degradation, and informative messages for users when issues occur. The deployment architecture shall allow the use of load balancers, replica sets, and failover mechanisms to maintain availability even during failures of individual components.

3.7.5 Usability

The system should be easy to learn and efficient to use for a wide range of users, including those with limited experience in online shopping. The interface shall follow established usability principles, use clear language, and provide helpful feedback. It shall be fully responsive and accessible on a variety of devices and screen sizes.

3.7.6 Maintainability

The system should be designed with maintainability in mind. Source code shall follow consistent naming conventions, be organized into clearly defined modules, and be supported by

documentation and tests. The architecture shall allow individual components to be updated or replaced without requiring changes to the rest of the system.

3.7.7 Compliance

The system shall comply with relevant legal and regulatory requirements, including data protection rules and consumer rights. It shall allow users to view the privacy policy, terms of service, and return policy. The system shall also support requests for data export or deletion where applicable.

3.7.8 Backup and Recovery

The system should support regular backups of the database and other critical data. Backups shall be stored in secure, redundant locations and shall be tested periodically for recoverability. Recovery procedures should be documented and practiced so that the system can be restored quickly after an incident.

3.7.9 Performance Monitoring and Logging

The system should include monitoring and logging facilities that provide visibility into its behaviors. Logs shall capture significant events, errors, and performance metrics. Alerts shall be configured for critical conditions, allowing administrators to respond quickly. Log data shall be retained for a sufficient period to support troubleshooting and auditing.

Chapter 4

DESIGN AND ARCHITECTURE

4. Design and Architecture

The design and architecture of the E-Commerce Website with AI-Powered Features are the result of a careful study of best practices in web engineering, software architecture, and machine learning integration. The design aims to deliver a platform that is both robust and flexible, capable of meeting current requirements while remaining open to future extension. This chapter presents the architectural style, components, data representation, process flows, and design models that together define the structure of the system.

4.1 System Architecture

The system architecture describes the overall structure of the platform, the main components that make it up, and the way in which they interact. The architecture is intended to provide a clear separation of concerns, to support scalability, and to simplify maintenance.

4.1.1 Architecture Style

The E-Commerce Website with AI-Powered Features follows a layered, service-oriented architecture built on top of the MERN stack. The architecture is divided into three main tiers: the presentation tier, implemented as a React-based single-page application; the application tier, implemented as a Node.js and Express.js backend that exposes REST APIs; and the data tier, implemented using MongoDB. This separation allows each tier to be developed, tested, deployed, and scaled independently. The style is augmented with modular services for authentication, recommendations, search, and chatbot functionality, each exposed through a consistent API.

4.1.2 Components

The system consists of several cohesive components, each responsible for a specific area of functionality. These components are described in the following sub-sections.

4.1.2.1 User Interface (UI)

The user interface is implemented as a React-based single-page application that delivers a rich, responsive, and interactive experience. It is organized into reusable components such as navigation bars, product cards, forms, dashboards, and modals. The UI communicates with the backend through REST API calls and uses client-side routing to provide a fast, app-like experience. Styling is handled using modern CSS frameworks and responsive design techniques that ensure consistency across devices.

4.1.2.2 Authentication Service

The authentication service handles user registration, login, password management, and session validation. It uses JWT-based authentication to issue tokens upon successful login, which are subsequently included in the headers of protected API requests. Passwords are stored using bcrypt hashing, and the service enforces password strength requirements. The authentication service also supports role-based access control, ensuring that only authorized users can access privileged endpoints.

4.1.2.3 Product Catalogue Module

The product catalogue module manages the creation, retrieval, update, and deletion of products and categories. It provides APIs for browsing, filtering, and searching the catalogue, as well as for administrative operations such as bulk import and export. The module interacts with the database to ensure that product information is consistently stored and efficiently retrieved.

4.1.2.4 Shopping Cart and Order Module

The shopping cart and order module handles all operations related to the shopping experience, including adding and removing items, updating quantities, applying coupons, and processing checkout. It is responsible for creating orders, updating inventory, and coordinating with the payment gateway. The module also manages order status updates and notifications to customers.

4.1.2.5 Payment Module

The payment module integrates with one or more payment gateways to process transactions securely. It abstracts the details of each gateway behind a consistent interface, allowing the system to support multiple payment providers without changes in the business logic. The module handles payment initiation, verification, and error recovery, and records all transactions in the database for auditing and reconciliation.

4.1.2.6 AI Services Module

The AI services module encapsulates the recommendation engine, smart search, and chatbot functionality. Each of these services is exposed through REST APIs that can be consumed by the frontend. The recommendation engine uses collaborative filtering, content-based filtering, and hybrid approaches to generate personalized suggestions. Smart search uses NLP techniques to interpret queries and rank results. The chatbot uses a combination of rule-based and machine-learning-based approaches to provide conversational assistance. The module is designed to be modular, allowing individual components to be updated or replaced without affecting the rest of the system.

4.1.2.7 Admin Dashboard

The admin dashboard provides tools for administrators to manage products, orders, customers, sellers, payments, and AI components. It also includes analytics views that present key performance indicators and other business metrics. The dashboard is implemented as part of the React application, with dedicated routes that are protected by role-based access control.

4.1.3 Interactions

The components interact with each other through well-defined interfaces. The front end calls the backend through REST APIs, passing JWT tokens in the authorization header. The backend calls internal services such as the authentication service, the product catalogue, and the AI services. The AI services interact with the database to read behavioral data and with the backend to return their results. The payment module interacts with external gateways and with the order module to coordinate transactions. All interactions are logged, monitored, and secured.

4.1.4 System Architecture Diagram

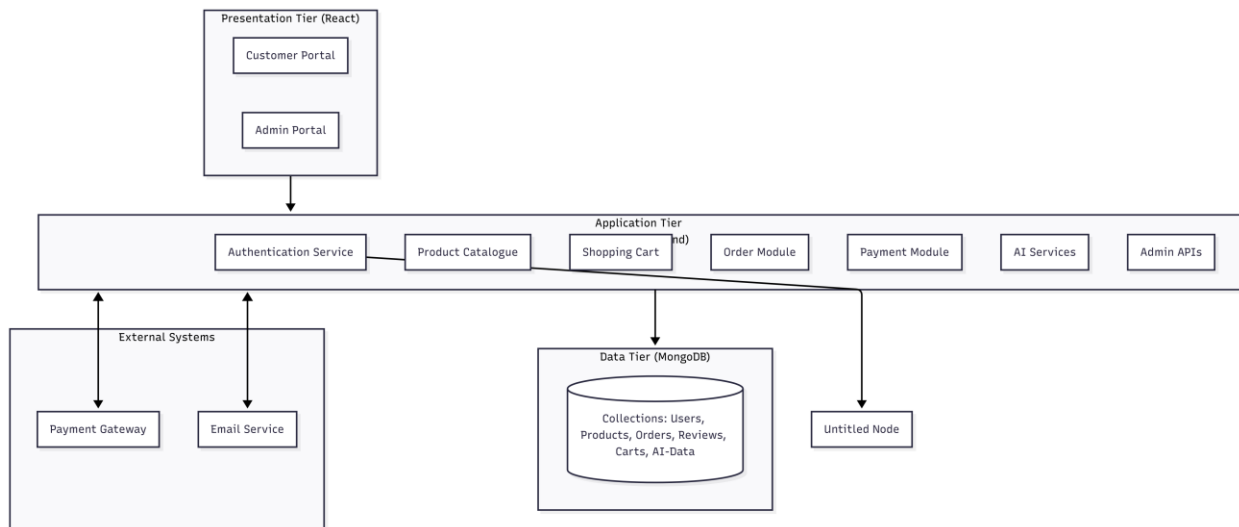


Figure 2: System Architecture Diagram

The system architecture diagram shows the three main tiers of architecture and the relationships between their components. The presentation tier at the top includes the React application, with separate sections for customers and administrators. The application tier in the middle includes the authentication service, product catalogue, shopping cart, order module, payment module, AI services, and admin APIs. The data tier at the bottom includes MongoDB collections for users, products, orders, reviews, carts, and AI-related data. External systems such as payment gateways and email services are connected to the application tier through dedicated integration points.

4.2 Data Representation

Data representation defines how information is stored, organized, and accessed within the system. Given the choice of MongoDB as the database, the representation is based on flexible, document-oriented schemas that accommodate the diverse data types handled by the platform.

4.2.1 Database Design

The database is structured as a set of collections, each corresponding to a logical entity of the system. The main collections include Users, Products, Categories, Carts, Orders, Payments, Reviews, Chats, and AI Models. Each collection stores documents with well-defined fields but allows additional fields to be added as requirements evolve. Indexes are created in frequently queried fields such as user email, product category, and order status to ensure fast retrieval. The scheme also considers reference and embedded data strategies, with user-specific data often embedded for fast access and catalogue data referenced to support consistency.

4.2.2 Entity-Relationship Representation

Although MongoDB is a document-oriented database, it is useful to describe the entities and their relationships in an entity-relationship style to clarify the design. Users are related to orders (one-to-many), orders are related to products through order items (many-to-many), carts are related to users and products, reviews are related to users and products, and AI models are related to users and products through behavioral data. The entity-relationship representation serves as a communication tool between designers and developers and guides the implementation of indexes and queries.

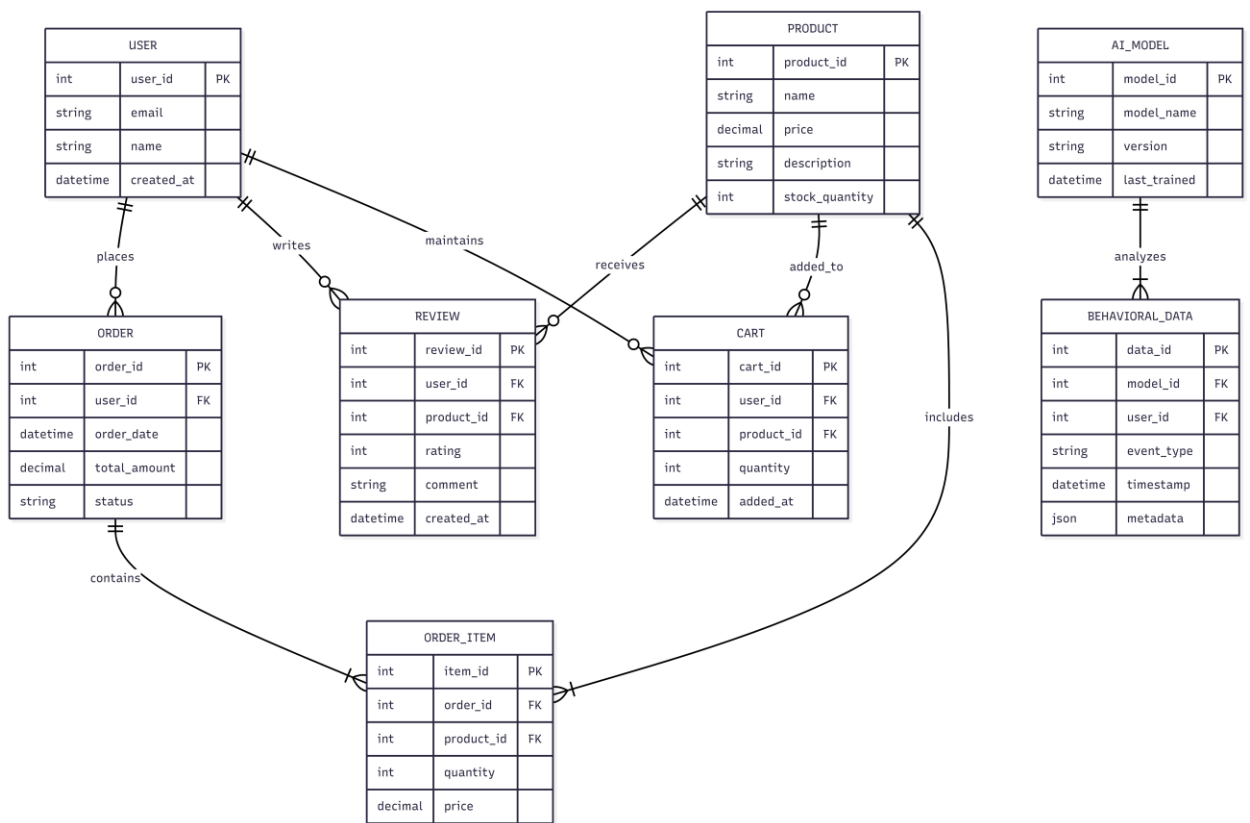


Figure 3:Entity-Relationship Diagram of the System

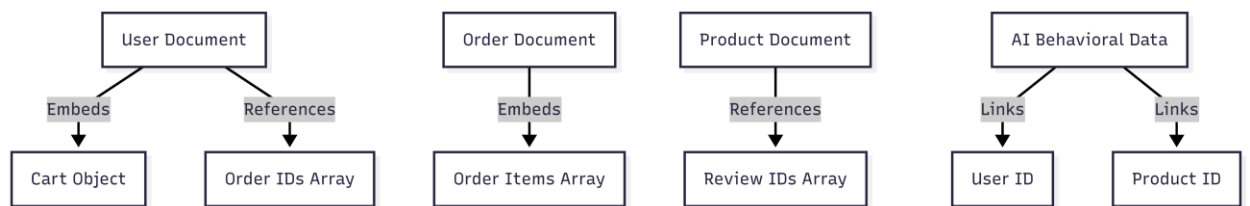


Figure 4:Detailed Document Relationship Structure in MongoDB

4.2.3 Data Flow

Data flows through the system in patterns that correspond to the main user workflows. When a customer registers, data flows from the UI to the authentication service and then to the Users collection. When a customer browses products, data flows from the Products collection through the catalogue module to the UI. When a customer places an order, data flows from the UI through the order module, payment module, and Orders collection, with updates to the Products collection for inventory. The AI services read from multiple collections to generate recommendations and process search queries and write back to dedicated analytical collections.

4.2.3.1 Sequence Diagram (Data Flow Across Layers)

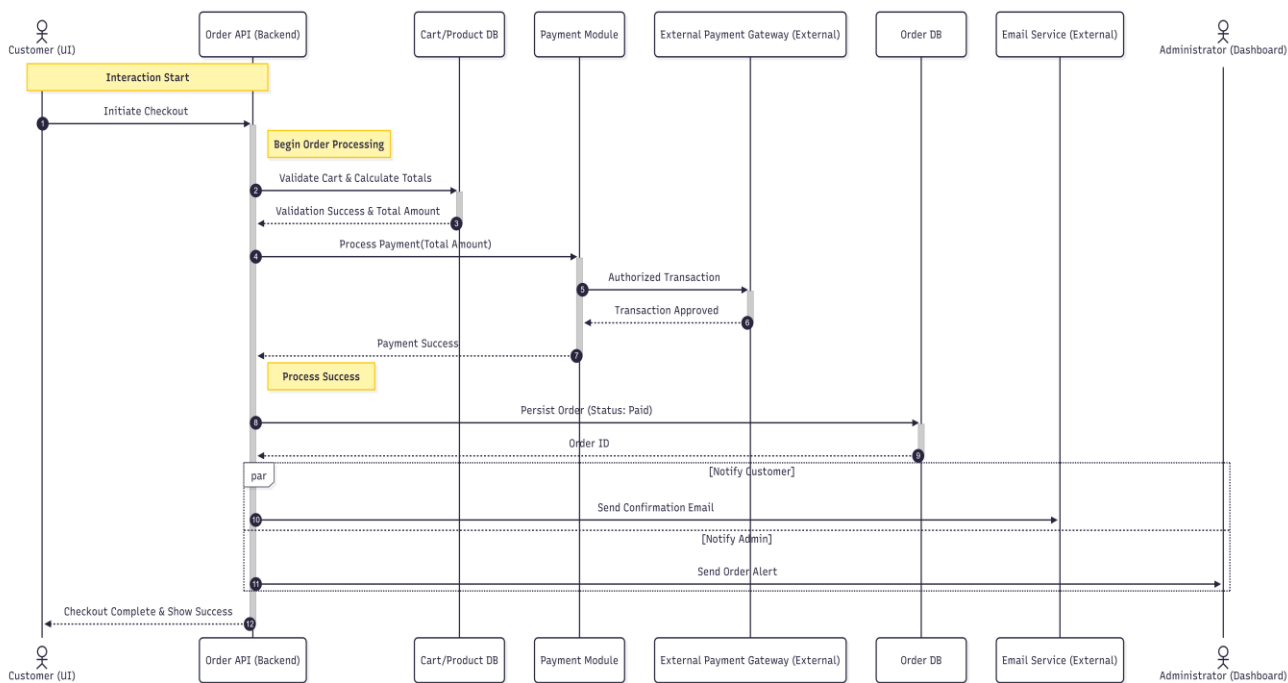


Figure 5:Sequence Diagram

The layered sequence diagram shows a typical interaction in which a customer places an order. The customer initiates the checkout from the UI, which calls the order API. The order API validates the cart, calculates totals, and invokes the payment module. The payment module interacts with an external gateway and returns the result. On success, the order persists and confirmation messages are sent to the customer and the administrator.

4.2.3.2 Activity Diagram (Overall System Workflow)

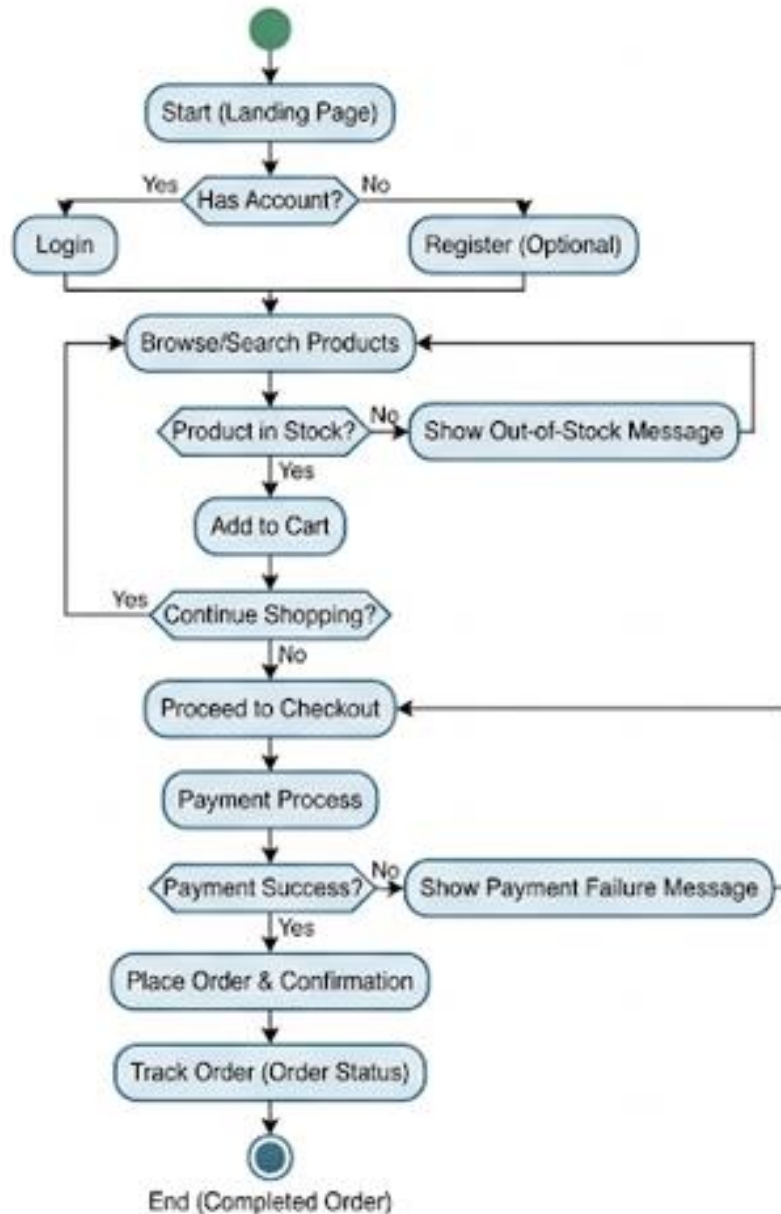


Figure 6: Vertical Customer-Journey Activity Diagram

The activity diagram summarizes the customer journey from initial visit to completed order. It starts with the landing page, proceeds through registration or login, moves to browsing or searching, continues with adding items to the cart, and ends with checkout, payment, and order tracking. Decision points along the way handle cases such as guest vs registered users, stock availability, and payment success or failure.

4.2.4 Entity Descriptions and Constraints

Each entity in the system is described with a clear set of attributes and constraints. User documents include fields such as name, email, password hash, role, and addresses, with email constrained to be unique. Product documents include name, description, category, brand, price, stock, and images, with category referenced from the Categories collection. Order documents include a reference to the user, an array of items, total amount, status, and timestamps. Constraints are enforced both at the application level and at the database level through validation rules.

4.3 Process Flow / Representation

This section describes the main process flows of the system in detail. Each process is broken down into steps that collectively define the expected behaviors.

4.3.1 Login and Authentication Process

The login process begins with the user entering credentials on the login page. The frontend sends the credentials to the authentication service. The service retrieves the user document from the database, verifies the password hash, and, if valid, generates a JWT token. The token is returned to the client and stored in a secure location. Subsequent API calls include the token in the authorization header, which is verified by the backend before processing the request.

4.3.2 Product Browsing and Search Process

When the user browses the catalogue, the frontend requests products from the catalogue API. The API queries the database with the appropriate filters and returns a paginated result. For search, the frontend sends the query to the smart search API, which uses NLP techniques to interpret the query, retrieves candidate products, ranks them using similarity and personalization signals, and returns the ranked list.

4.3.3 Recommendation Generation Process

The recommendation process is triggered when the user opens the home page, a product detail page, or the cart. The frontend requests recommendations from the AI services. The service retrieves the user's behavioral data and the product catalogue, applies the configured recommendation strategies, and returns a ranked list of suggestions. The results are displayed in dedicated sections of the UI, such as "Recommended for you" or "Customers also viewed".

4.3.4 Payment Processing Workflow

Payment processing starts when the user confirms the checkout. The order module creates a pending order and invokes the payment module, which contacts the external gateway. The gateway returns a success or failure response. On success, the order is marked as paid, inventory is updated, and confirmation emails are sent. On failure, the user is notified and the order is either cancelled or retried, depending on the user's action.

4.4 Design Models

Several UML models are used to capture different aspects of the design. Each model provides a specific perspective on the system and supports the implementation phase.

4.4.1 Class Model Description

The class model represents the main classes in the system, their attributes, and their methods. Key classes include User, Product, Category, Cart, Order, Order Item, Payment, Review, Chat Message, and Recommendation. Each class encapsulates a specific set of responsibilities and collaborates with other classes through well-defined interfaces. Relationships such as inheritance, association, and aggregation are used to model shared attributes and behaviors.

4.4.2 Sequence Model Description

The sequence model shows the order of interactions between objects for typical scenarios such as login, order placement, and recommendation retrieval. It highlights the sequence of method calls, the passing of parameters, and the return of results, giving a clear picture of the runtime behaviors of the system.

4.4.3 Activity Model Description

The activity model represents the overall workflow of the system as a series of activities and decision points. It complements the sequence model by focusing on the business-level flow rather than the technical sequence. Activities include browsing, searching, adding to cart, checking out, and receiving recommendations.

4.4.4 Component Model Description

The component model describes the system at the architectural level, showing the main components and their dependencies. It groups related classes into components such as UI, Authentication, Catalogue, Cart, Order, Payment, AI Services, and Admin. The component model is particularly useful for understanding deployment and scaling considerations, and for identifying the boundaries between modules.

Chapter 5

IMPLEMENTATION

5. Implementation

The implementation phase of the E-Commerce Website with AI-Powered Features translates the design into working software. This chapter describes the algorithms used for the key functionalities of the system, the external APIs that the platform consumes, and the user interface screens that deliver the final experience to customers and administrators. The implementation follows the design principles and architectural guidelines established in the previous chapters, ensuring that the resulting system is aligned with the intended vision of the project.

5.1 Algorithms

Several algorithms are implemented to support the core features of the system. Each algorithm is described in a consistent format that explains its purpose, input, output, and main steps.

5.1.1 Authentication Algorithm

The authentication algorithm validates user credentials and issues JWT tokens for authenticated users. It is invoked during both registration and login. During registration, the algorithm generates a bcrypt hash of the password with a configurable cost factor and stores the hash in the Users collection along with the user's other details. During login, the algorithm retrieves the user document by email, compares the provided password with the stored hash using bcrypt, and, on success, generates a JWT token signed with a secret key. The token contains the user ID, role, and expiry time, and is returned to the client. Token verification is performed on protected endpoints by decoding the JWT and validating its signature and expiry. If any step fails, a corresponding error is returned to the client.

5.1.2 Recommendation Algorithm

The recommendation algorithm generates personalized product suggestions for each user. It uses a hybrid approach that combines collaborative filtering and content-based filtering. Collaborative filtering identifies users with similar behavior patterns and recommends products that these similar users have interacted with. Content-based filtering analyses the attributes of products the user has already viewed or purchased and recommends similar products based on category, brand, price range, and other features. The results of both approaches are combined using weighted scoring, with weights adjusted based on data availability and user activity. The algorithm returns a ranked list of recommended products, with explanations that can be optionally displayed to the user.

5.1.3 Smart Search Algorithm

The smart search algorithm interprets natural language queries and returns relevant products. It starts by tokenizing the query, removing stop words, and applying stemming. Spelling correction is then applied using an edit-distance approach combined with a dictionary of product-related terms. Synonym expansion is performed using a curated thesaurus and, optionally, a learned embedding model. The expanded query is matched against the product index, and candidate products are scored based on term frequency, inverse document frequency, and additional signals

such as popularity and personalization. The algorithm returns to a ranked list of products, optionally accompanied by related search suggestions.

5.1.4 Chatbot Algorithm

The chatbot algorithm handles conversational interactions with customers. It combines rule-based pattern matching common queries with machine-learning-based intent recognition for more complex questions. The input message is preprocessed, classified into an intent category, and mapped to a response template or action. For actions that require dynamic data, such as order status, the algorithm calls the appropriate internal API and composes a response using the result. The algorithm also maintains a short conversational memory to support multi-turn interactions. When confidence in the intent is low, the algorithm hands the conversation over to a human agent with a concise summary of the interaction.

5.1.5 Payment Processing Algorithm

The payment processing algorithm coordinates the interaction with external payment gateways. It prepares a payment request containing the order total, currency, and customer information, and sends it to the gateway through a secure API call. The algorithm handles various outcomes, including success, failure, and pending states, and updates the order status accordingly. For pending payments, the algorithm periodically polls or listens for webhook notifications from the gateway. In case of failure, the algorithm offers the customer the option to retry the payment or select an alternative method.

5.2 External APIs

The platform integrates with several external APIs to deliver its functionality. These APIs are selected based on reliability, quality, and relevance to the needs of the system. The following table summarizes the main APIs used in the project.

Table 16:Details of APIs Used in the Project

API	Purpose	Provider	Usage
Stripe / PayPal	Payment processing	Stripe / PayPal	Secure card and wallet payments at checkout
OpenAI / Hugging Face	NLP for chatbot and search	OpenAI / Hugging Face	Intent recognition, embeddings, language understanding
Google Maps	Address validation	Google	Shipping address autocompletes and validation
SendGrid / SMTP	Email notifications	SendGrid	Order confirmation and password reset emails

Cloud nary	Image hosting	Cloud nary	Storing and serving product images
MongoDB Atlas	Managed database	MongoDB	Cloud-hosted MongoDB cluster
Firebase	Push notifications	Google	Real-time notifications to customers
Vercel / Render	Deployment	Vercel / Render	Hosting of frontend and backend

Each external API is consumed through a dedicated wrapper module in the backend, which isolates the rest of the system from provider-specific details. This design allows providers to be swapped with minimal disruption and supports environment-based configuration for development, testing, and production.

5.3 User Interface and Screenshots

The user interface of the E-Commerce Website with AI-Powered Features is the most visible outcome of the implementation phase. It is designed to be clean, modern, accessible, and consistent with the expectations of contemporary online shoppers. The interface follows a component-based structure, uses a unified design system, and has been validated through manual usability sessions. This section documents the final deployed screens of the platform. Each sub-section contains an academic description of the screen, a placeholder for the corresponding screenshot, a figure caption, and a cross-reference that is used in the running text of the report.

5.3.1 Home Page

The home page is the first screen that users encounter when they visit the platform. It presents the brand identity, featured categories, promotional banners, and a dedicated section for personalized product recommendations generated by the AI engine. The top navigation bar contains links to the main sections of the store, the intelligent search bar, a cart icon with a live item counter, and an account icon that switches between login and profile entries based on the authentication state.

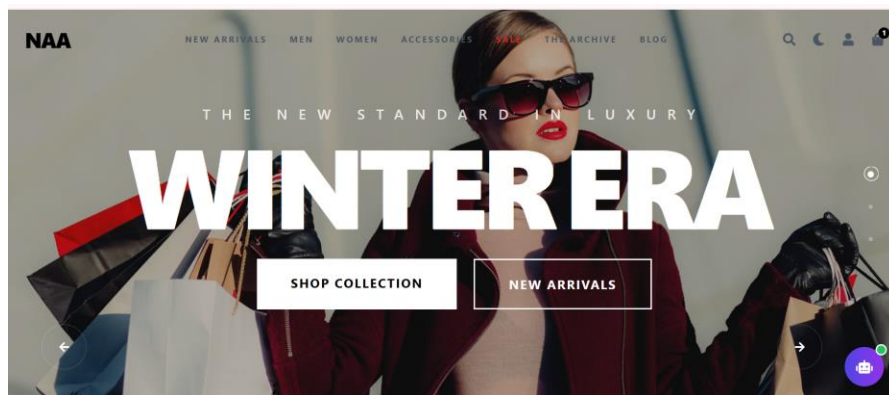


Figure 7: Home Page of E-Commerce Website

As shown in Figure 5.1, the homepage displays the hero banner, featured product carousels, personalized recommendations, and the primary navigation that leads customers toward the catalogue and account areas of the platform.

5.3.2 Login / Register Page

The login and registration page allows new users to create an account and returning users to sign in securely. The interface presents a clearly labelled form with validated fields for name, email, and password, and enforces password strength requirements. Helpful feedback messages are displayed for invalid inputs, duplicate registrations, or incorrect credentials, and a link is provided for password recovery.

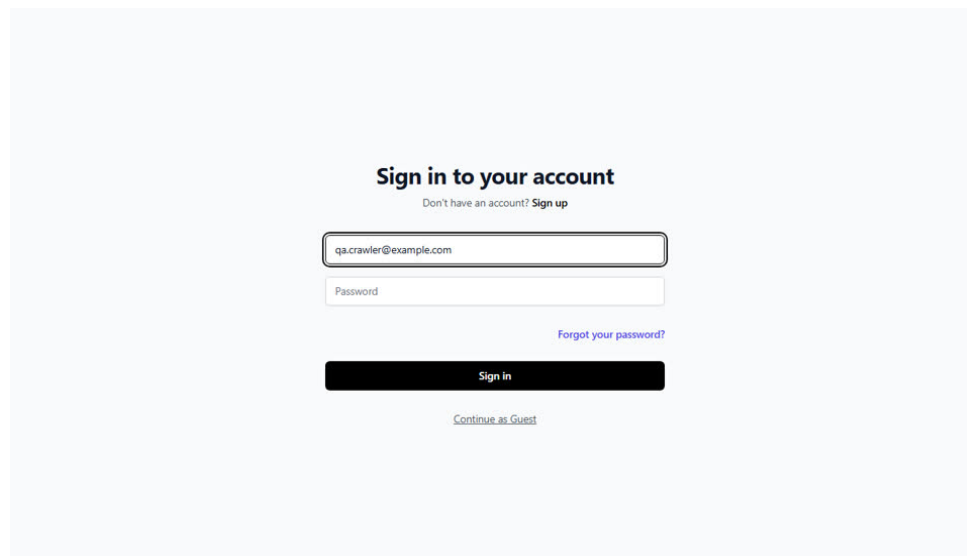


Figure 8: Login / Register Page of E-Commerce Website

As shown in Figure 8, the login and registration page provides a unified, form-based entry point where users can switch between creating a new account and signing in, with inline validation and clear error messaging.

5.3.3 Product Listing Page

The product listing page displays the products that belong to a selected category or that match a search query. Each product is represented by a consistent card that shows the image, name, price, discount label, average rating, and quick action controls. A sidebar and a top toolbar together provide filtering and sorting options such as price range, brand, and rating, while pagination supports efficient navigation through long catalogues.

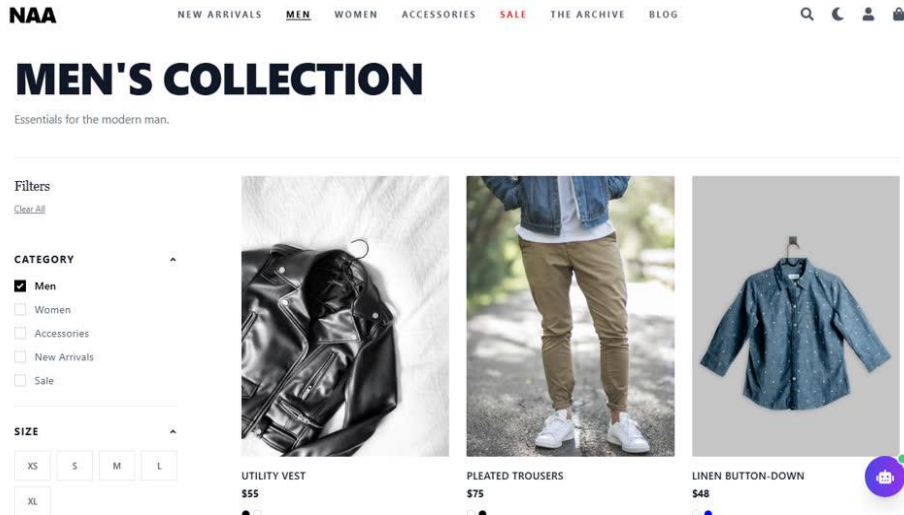


Figure 9:Product Listing Page of E-Commerce Website

As shown in Figure 9, the product listing page renders the available products in a responsive grid of cards, accompanied by filtering and sorting controls that allow customers to narrow down the selection according to their preferences.

5.3.4 Product Detail Page

The product detail page provides comprehensive information about a specific product. It includes a gallery of high-resolution images, a detailed textual description, technical specifications, pricing, stock status, variant selection, customer reviews, and a prominent "Add to Cart" action. A dedicated section below the main content displays related products suggested by the recommendation engine, supporting further discovery.

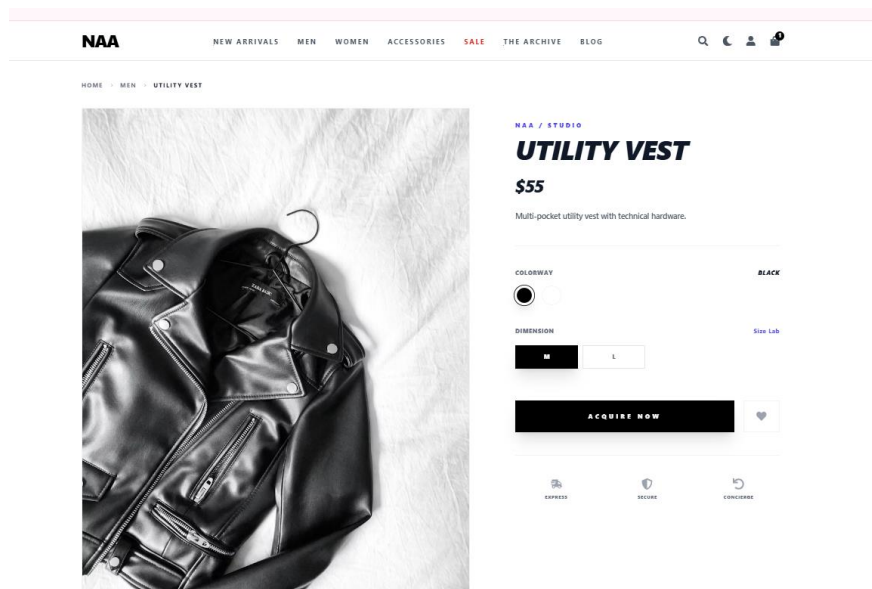


Figure 10:Product Detail Page of E-Commerce Website

As shown in Figure 10, the product detail page consolidates all relevant product information into a single view, while recommended alternatives appear beneath it to support informed purchasing decisions.

5.3.5 Shopping Cart Page

The shopping cart page lists every item currently added to the cart, with controls for updating quantities, removing individual items, and applying coupon codes. The page recalculates the subtotal, applied discounts, and estimated shipping in real time as the contents change, giving the customer a clear overview of the pending purchase before proceeding to check out.

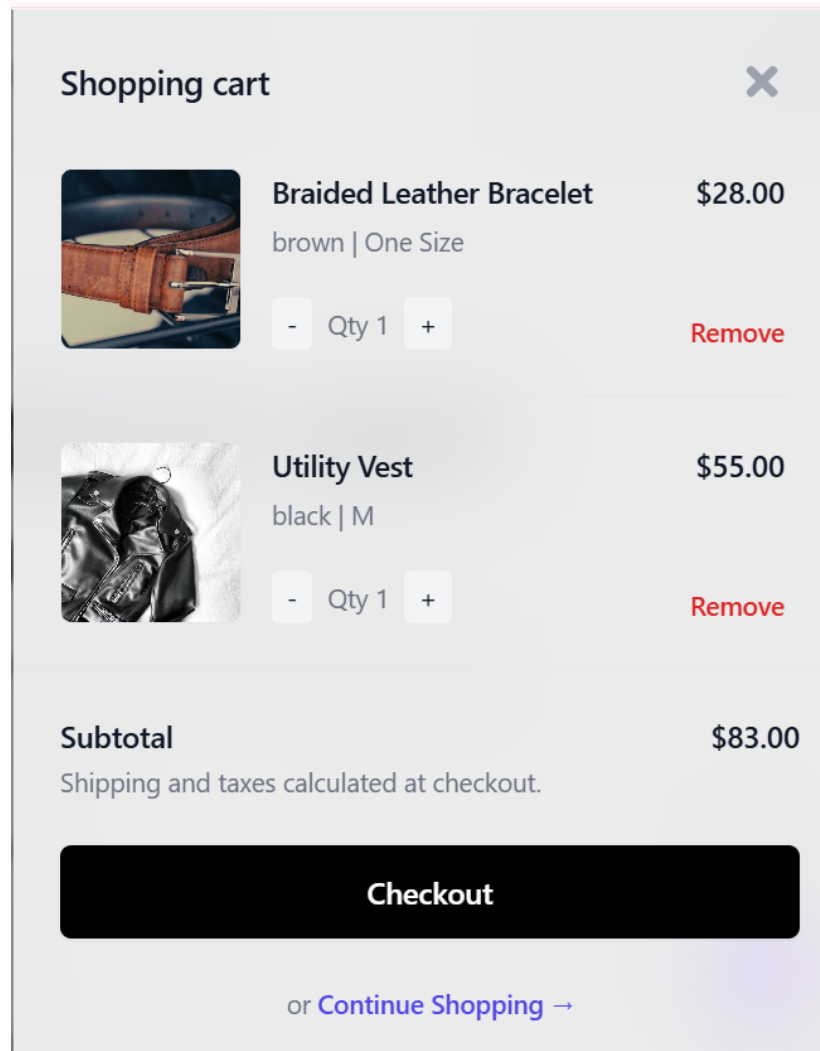


Figure 11: Shopping Cart Page of E-Commerce Website

As shown in Figure 11, the shopping cart page presents the selected items in a structured list, summarizes the totals, and offers a clear call to action that leads the customer to the checkout flow.

5.3.6 Checkout Page

The checkout page collects shipping and billing information, displays the final order summary, and allows the customer to select a payment method such as credit or debit card, a digital wallet, or cash on delivery where available. The flow is structured into well-delimited steps, includes client-side validation, and integrates securely with the external payment gateway.

NAA NEW ARRIVALS MEN WOMEN ACCESSORIES SALE THE ARCHIVE BLOG

Checkout

Contact Information

Email address

Shipping Address

First name Last name

Address

City ZIP Code

Payment

Card

Secure, fast checkout with Link

Card number

1234 1234 1234 1234

Expiration date

MM / YY

Security code

CVC

Country

Pakistan

Order Summary

Braided Leather Bracelet	\$28.00
Utility Vest	\$55.00
Subtotal	\$83.00
Shipping	\$10.00
Total	\$93.00

Pay \$93.00

Figure 12:Checkout Page of E-Commerce Website

As shown in Figure 12, the checkout page guides the customer through address entry, payment method selection, and final order confirmation, preserving security and clarity throughout the process.

5.3.7 Admin Dashboard

The admin dashboard presents a high-level overview of the business to administrators. Key performance indicators such as total sales, total orders, registered users, and conversion rate are displayed at the top of the page through cards and charts. A side navigation panel provides access to management screens for products, orders, users, sellers, and analytics, supporting the day-to-day operation of the platform.

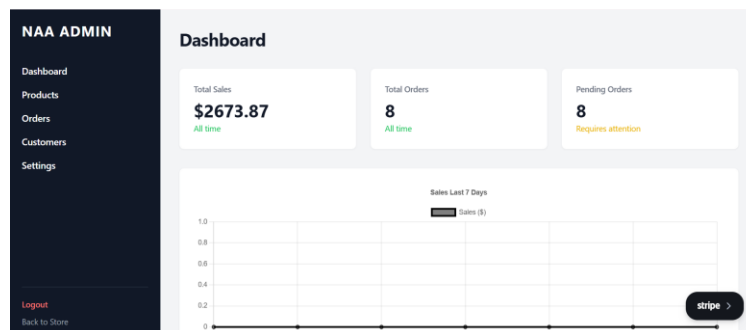


Figure 13:Admin Dashboard of E-Commerce Website

As shown in Figure 13, the admin dashboard summarizes the most important business metrics briefly and offers a single navigation surface for all administrative operations.

5.3.8 Admin — Product Management

The product management screen allows administrators to create, update, and remove products and categories. Each product can be configured with images, descriptions, pricing, inventory, variants, and AI-related tags. The interface supports search, filtering, and bulk operations, enabling the catalogue to be maintained efficiently even as it grows.

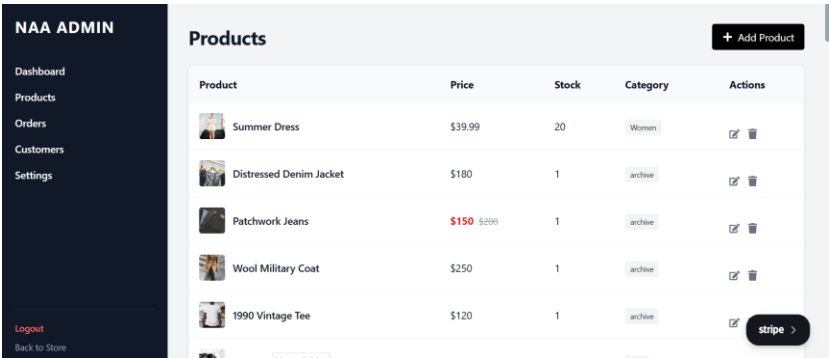


Figure 14:Admin Product Management of E-Commerce Website

As shown in Figure 5.8, the admin product management screen provides a tabular interface for maintaining the catalogue, complete with inline actions and forms for adding or editing product records.

5.3.9 Admin — Orders Management

The orders management screen enables administrators to view and process customer orders. Orders are presented with their status, payment method, total amount, and customer information. Administrators can update the status, issue refunds, and export order data for accounting and analytics purposes.

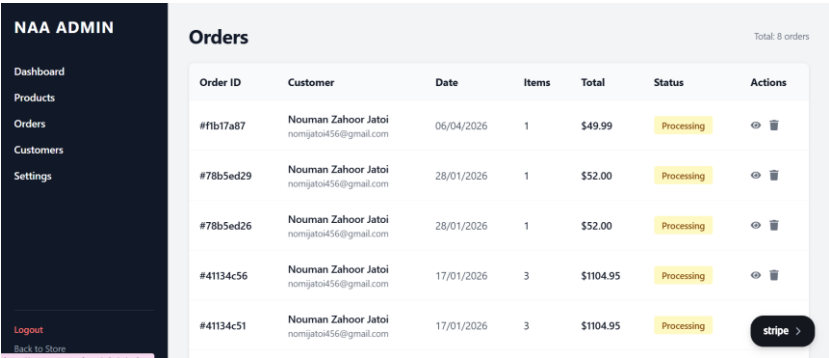


Figure 15:Admin Orders Management of E-Commerce Website

As shown in Figure 15, the admin orders management screen consolidates all incoming orders into a filterable table and supports status transitions and follow-up actions required for order fulfilment.

Collectively, the screens documented above demonstrate that the implementation fulfils the requirements captured in Chapter 3 and adheres to the design principles established in Chapter 4. The consistent use of a shared design system, responsive layouts, and accessible interaction patterns ensures that the platform is usable across a variety of devices and contexts.

Chapter 6

TESTING AND EVALUATION

6. Testing and Evaluation

Testing and evaluation are essential activities that verify the quality, correctness, and reliability of the E-Commerce Website with AI-Powered Features. This chapter describes the testing strategy adopted for the project, including manual and automated approaches, and presents the results of representative test cases in structured tables. The testing strategy covers multiple levels, from individual functions to full end-to-end scenarios, and considers both functional and non-functional aspects of the system.

6.1 Manual Testing

Manual testing plays an important role in verifying the behavior of the system, especially for aspects that are difficult to automate, such as user experience, visual appearance, and exploratory interactions. Manual tests are designed based on the functional requirements, use cases, and design specifications, and are executed by the development team as well as selected users who act as testers.

6.1.1 System Testing

System testing verifies that the integrated system meets the specified requirements. It is performed after all the main modules have been implemented and integrated. System tests are designed around real-world scenarios, such as customer registering, browsing, searching, adding items to a cart, paying, and tracking the order. Similar scenarios are defined for administrators, including product management, order management, and analytics.

6.1.1.1 Purpose of System Testing

The purpose of system testing is to validate that the complete platform behaves as expected under representative conditions. System testing focuses on end-to-end flows and ensures that the integration of individual modules does not introduce inconsistencies or regressions.

6.1.1.2 Types of System Testing

Several types of system testing are applied, including functional testing, usability testing, performance testing, security testing, and compatibility testing. Each type addresses a different aspect of the system and contributes to a comprehensive understanding of its quality.

6.1.1.3 Importance of Manual System Testing

Manual system testing is important because it allows testers to evaluate aspects that automated tests cannot easily verify, such as the aesthetics of the UI, the clarity of messages, and the naturalness of conversational flows. It also enables exploratory testing, where testers deviate from predefined scripts to uncover unexpected issues.

6.1.2 Unit Testing

Unit testing focuses on individual units of code, such as functions or methods, and verifies that each unit behaves correctly in isolation. Unit tests are typically written alongside the production code and are executed frequently during development to catch regressions early.

Table 17:Unit Testing Table

Test ID	Module	Description	Expected Result	Status
UT-01	Authentication	Hash password with crypt	Hash generated and matched on verification	Pass
UT-02	Authentication	Issue JWT token on valid login	Valid token returned with correct payload	Pass
UT-03	Catalogue	Create product with valid data	Product created and stored in DB	Pass
UT-04	Catalogue	Reject product with missing fields	Validation error returned	Pass
UT-05	Cart	Add item to cart	Cart updated with new item	Pass
UT-06	Cart	Reject negative quantity	Validation error returned	Pass
UT-07	Order	Calculate order total with coupon	Total reduced by coupon amount	Pass
UT-08	Recommendation	Return empty list for new user	Fallback to popular products	Pass
UT-09	Search	Correct common misspelling	Corrected query used for matching	Pass
UT-10	Chatbot	Respond to greeting	Friendly greeting response returned	Pass

6.1.3 Functional Testing

Functional testing verifies that the system behaves as specified for each user's story or feature. Functional tests are organized around use cases and describe concrete inputs, actions, and expected outputs.

Table 18:Functional Testing Table

Test ID	Feature	Description	Expected Result	Status
FT-01	Registration	New user registers with valid data	Account created, confirmation shown	Pass
FT-02	Login	User logs in with correct credentials	User redirected to home, token issued	Pass
FT-03	Login	User logs in with wrong password	Error message displayed	Pass
FT-04	Browse	User filters products by category	Only matching products shown	Pass
FT-05	Smart Search	User searches with misspelled query	Correct products returned	Pass
FT-06	Cart	User updates cart quantity	Cart total recalculated	Pass
FT-07	Checkout	User completes payment	Order created, confirmation email sent	Pass
FT-08	Recommendation	Returning user views home page	Personalized recommendations shown	Pass
FT-09	Chatbot	User asks about order status	Bot retrieves status and responds	Pass
FT-10	Admin	Admin adds a new product	Product visible in catalogue	Pass
FT-11	Admin	Admin updates order status	Customer notified of new status	Pass
FT-12	Admin	Admin views analytics	Charts rendered correctly	Pass

6.1.4 Integration Testing

Integration testing verifies that different modules of the system work correctly together. It focuses on the interactions between the frontend and backend, between the backend and the database, and between the backend and external services such as payment gateways.

Table 19: Integration Testing Table

Test ID	Modules	Description	Expected Result	Status
IT-01	Frontend<-Backend	API call for product list	Backend returns paginated products	Pass
IT-02	Backend<-> MongoDB	Persist new user document	Document saved and retrievable	Pass
IT-03	Backend<-> Gateway	Process test payment	Payment approved by sandbox	Pass
IT-04	Backend<-> Email	Send registration email	Email delivered to test inbox	Pass
IT-05	Backend <-> AI Service	Call recommendation API	Valid recommendations returned	Pass
IT-06	Frontend <-> AI Chatbot	Send chat query and display response	Response rendered in chat window	Pass
IT-07	Admin <-> Backend	Bulk update product inventory	All products updated correctly	Pass
IT-08	Search <-> Catalogue	Smart search queries the catalogue	Relevant products ranked	Pass

6.2 Automated Testing

Automated testing complements manual testing by providing fast, repeatable checks that can be run as part of the development workflow. Automated tests are particularly useful for regression testing, where the goal is to ensure that new changes do not break existing functionality.

The project uses a combination of unit tests, API tests, and end-to-end tests. Unit tests are written using a framework such as Jest and executed on every commit. API tests are written using supertest or a similar library and verify the behavior of REST endpoints. End-to-end tests are written using a framework such as Cypress and simulate real user interactions in a browser environment.

Table 20:Automated Testing Table

Test ID	Type	Description	Framework	Status
AT-01	Unit	Password hashing and verification	Jest	Pass
AT-02	Unit	Cart total calculation	Jest	Pass
AT-03	API	POST /api/auth/register	Supertest	Pass
AT-04	API	POST /api/auth/login	Supertest	Pass
AT-05	API	GET /api/products? Category=electronics	Supertest	Pass
AT-06	API	POST /api/orders	Supertest	Pass
AT-07	E2E	Customer registration and login flow	Cypress	Pass
AT-08	E2E	Browse, add to cart, and checkout	Cypress	Pass
AT-09	E2E	Smart search and open product	Cypress	Pass
AT-10	E2E	Chatbot interaction with order lookup	Cypress	Pass

Automated tests are integrated with the version control and continuous integration system so that they are executed on each change. Failed tests are reported back to the developer, who is expected to fix the underlying issue before merging the change. This discipline keeps the main branch of the repository in a healthy state and ensures that the team can release updates frequently without introducing regressions.

In addition to functional tests, the project uses performance tests to measure the behavior of the system under load. Tools such as Apache JMeter or k6 are used to simulate many concurrent users and to measure response times, throughput, and error rates. Performance tests help to identify bottlenecks and to validate that the system meets the performance targets defined in the non-functional requirements.

Security tests are also included in the overall testing strategy. They include checks for authentication bypass, authorization leaks, injection attacks, insecure direct object references, and exposure of sensitive data. Some of these checks are automated with tools such as OWASP ZAP, while others are performed manually during security reviews.

6.3 Visual Test Evidence (Screenshots)

This section presents visual evidence captured during the execution of the test cases described earlier in this chapter. Each screenshot corresponds to a representative scenario that confirms the correct behavior of a specific feature. For every scenario, the text provides a short academic description of what is being demonstrated, followed by a placeholder for the screenshot and a figure caption. In the running narrative, each screenshot is referenced explicitly through the in-text citation pattern required by the report style.

6.3.1 Successful Login

This test scenario verifies that a registered customer can authenticate successfully using valid credentials. After submitting the login form, the system validates the input, issues a JWT token, and redirects the user to the home page or the previously requested page. A personalized welcome indicator in the navigation bar confirms the authenticated state.

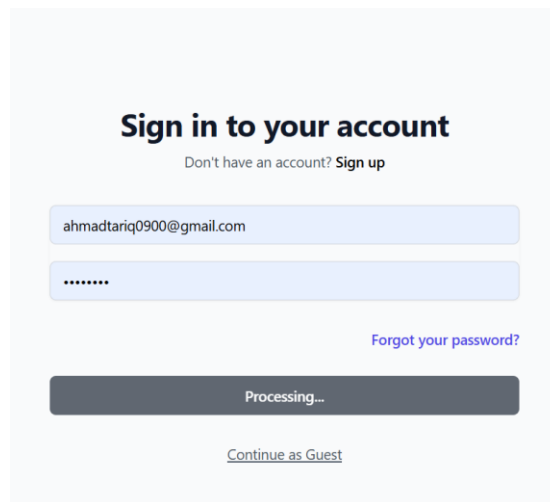


Figure 16: Successful Login of E-Commerce Website

As shown in Figure 16, the system correctly authenticates the user with valid credentials and transitions the interface to the logged-in state, demonstrating that the authentication workflow behaves as specified in the requirements.

6.3.2 Add to Cart

This test scenario verifies that a product can be added to the shopping cart from the product detail page. Once the user clicks the "Add to Cart" button, the cart counter in the navigation bar increases, a non-intrusive confirmation notification is shown, and the cart summary reflects the added item with its price, quantity, and selected variant.

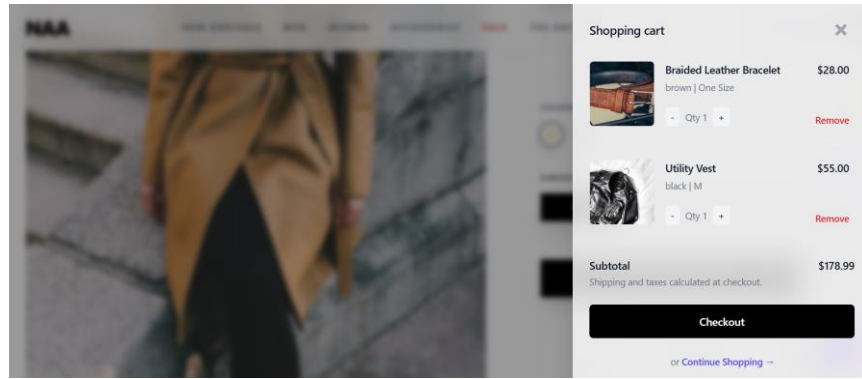


Figure 17: Add to Cart Functionality of E-Commerce Website

As shown in Figure 17, the system successfully registers the new item in the cart and updates the relevant UI indicators, confirming that the add-to-cart pathway operates correctly under the conditions defined by the test case.

6.3.3 Order Placement

This test scenario verifies the end-to-end order placement workflow. After proceeding from the cart to checkout, the user confirms the shipping address, selects a payment method, and completes the transaction through the integrated gateway. Upon success, the system creates the order, updates inventory, dispatches a confirmation email, and displays a confirmation page with the order number and a summary of the purchase.

6.3.4 Error Handling

This test scenario verifies the behavior of the system under invalid or unexpected conditions, including incorrect login credentials, attempts to check out with an empty cart, and failures returned by the payment gateway. The system responds with clear, user-friendly error messages, preserves the user's progress wherever possible, and guides the customer toward a corrective action.

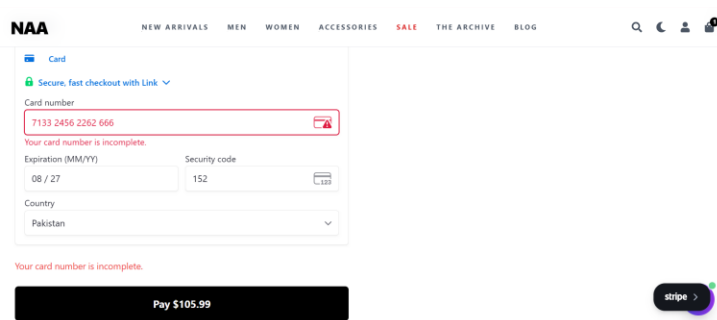


Figure 18: Error Handling Screens of E-Commerce Website

As shown in Figure 18, the platform handles erroneous situations gracefully by displaying informative messages instead of allowing the user to proceed with invalid state, confirming that the robustness requirements identified in Chapter 3 are satisfied.

Chapter 7

CONCLUSION

7. Conclusion

The E-Commerce Website with AI-Powered Features represents a comprehensive effort to build a modern, intelligent, and user-centric online shopping platform that combines proven web engineering practices with emerging artificial intelligence capabilities. Through careful requirement analysis, systematic design, disciplined implementation, and thorough testing, the project has delivered a fully functional system that demonstrates how AI can be embedded within a traditional e-commerce workflow to enhance the experience of both customers and administrators.

The platform demonstrates a clear advancement over typical local e-commerce solutions in several ways. It introduces a personalized recommendation engine that tailors product suggestions to each user, an intelligent search module that understands natural language queries, and a conversational chatbot that provides immediate assistance around the clock. These AI-driven capabilities are combined with strong foundations in authentication, cart management, checkout, payment processing, and administrative control, resulting in a system that is both feature-rich and well-engineered.

Beyond its immediate functional capabilities, the project also demonstrates the practical application of key concepts from the Bachelor of Science in Computer Science curriculum, including software engineering, database systems, web engineering, object-oriented programming, data structures, human computer interaction, information security, and artificial intelligence. The integration of these disciplines into a single project showcases the team's ability to bridge the gap between theoretical knowledge and practical delivery.

The project also highlights several lessons learned during its development. The integration of AI components requires careful attention to data quality, model lifecycle management, and the clear separation between intelligent features and core business logic. Security and privacy considerations must be integrated from the beginning rather than added at the end. Performance optimization and scalability planning, while sometimes considered secondary in academic projects, are essential for platforms intended to handle real-world traffic. Above all, the importance of iterative development, continuous feedback, and disciplined testing has been reinforced throughout the project.

In conclusion, the E-Commerce Website with AI-Powered Features delivers a cohesive, intelligent, and extensible platform that can serve as the foundation for a wide variety of digital retail scenarios. It addresses the specific shortcomings observed in local online stores, demonstrates the relevance of AI in e-commerce, and lays the groundwork for future expansion. The project has successfully met the objectives set at the beginning and provides a strong base for the future work described in the following section.

7.1 Future Work

Although the current version of the platform delivers complete shopping experience, several extensions and enhancements have been identified for future work. These are described in the following sub-sections.

7.1.1 Mobile Application

A native mobile application, developed in React Native or a similar framework, can complement the web platform and reach customers who prefer to shop from their smartphones. The mobile application would leverage the existing REST APIs and AI services and offer additional features such as push notifications, biometric authentication, and offline browsing of recently viewed products.

7.1.2 Multi-Vendor Marketplace

Expanding the platform into a full multi-vendor marketplace would allow multiple sellers to list their products and manage their own stores under a unified brand. Features required for this extension include seller onboarding, commission management, separate analytics for each seller, and enhanced dispute resolution. The current architecture is designed to accommodate this expansion with minimal redesign.

7.1.3 Subscription and Loyalty Programs

Subscription-based pricing models, loyalty programs, and membership tiers can further enhance customer retention and revenue. These features would allow customers to subscribe to regular deliveries of consumables, earn points on purchases, and enjoy exclusive benefits. Implementing such features requires extensions to the order model, the payment system, and the analytics module.

7.1.4 Advanced Analytics and Reporting

The analytics capabilities of the platform can be significantly expanded through the integration of advanced reporting and data visualization tools. Features such as cohort analysis, customer lifetime value, predictive demand forecasting, and anomaly detection would empower administrators to make more informed decisions. Integration with dedicated business intelligence platforms is also possible.

7.1.5 Integration with Voice Assistants and AR

Voice-based shopping and augmented reality product previews represent emerging trends that can further elevate customer experience. Voice integration would allow customers to search, add to cart, and place orders through voice commands, while AR-based previews would allow them to visualize products in their own environment before purchasing. These features require additional investment in front-end capabilities and integration with third-party SDKs.

7.1.6 Enhanced Security Features

Security is an area where continuous improvement is always warranted. Future enhancements include multi-factor authentication, advanced fraud detection using machine learning, security auditing tools, and compliance with additional standards such as PCI DSS and local data protection regulations. The platform is designed to accommodate these improvements as part of its ongoing evolution.

Chapter 8

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